21-602: Introduction to Set Theory I

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Chapter 1

A Recap of Undergraduate Set Theory

Set Theory can be viewed as a field of mathematics that encompasses all other mathematics. We can view set theory as beginning at the empty set $\emptyset$, also denoted V_0. We can construct subsequent sets $V_1, V_2, V_3, \ldots$ by taking power sets, ie, $V_1 = \mathcal{P}(V_0)$, $V_2 = \mathcal{P}(V_1)$, and so on. We can then define V_ω to be the union of all V_n, where $n < \omega$. We can then define $V_{\omega+1} = \mathcal{P}(V_\omega)$, and so on. All of analysis, the field of mathematics Davoud Cheraghi once described as the "rigorous study of infinite constructions", is essentially about $V_{\omega+1}$. In this way, set theory encompasses analysis, for example.

We call the hierarchy

$$V_0 \subset V_1 \subset \cdots \subset V_\omega \subset V_{\omega+1} \subset \cdots$$

the **universe of sets**. Indeed, this is often represented diagrammatically as a V that contains each of these Vs, with V_0 at the bottom, V_1 above V_0, and so on.

There is a problem with the universe of sets: *the power set operation is too strong*. That is, when we take the power set of a set, it "throws too many elements in". This leads to questions like the *continuum hypothesis*, which asks the question of whether $2^{\aleph_0} = |\mathbb{R}| = |V_{\omega+1}|$ is equal to $\aleph_1$.

Something we will study quite closely in this course is a slightly different version of this universe, known as **Gödel's Constructible Universe**. This is a universe of sets that is not built up from $\emptyset$ using the power set, but using a *different* operation $\mathcal{D}$, defined as follows: for a set A, viewing

it as a first-order structure $(A; \in)$,

$$\mathcal{D}(A) := \{x \subseteq A \mid x \text{ is 1st-order definable over } (A; \in) \text{ using parameters from } A\}$$

We can show that for finite sets, $\mathcal{D}$ agrees with $\mathcal{P}$. Moreover, we can show that for a countable set A, $\mathcal{D}(A)$ is countable as well. The point is that if we then define a universe

$$(L_0 = \emptyset) \subset (L_1 = \mathcal{D}(L_0)) \subset \cdots \subset \left(L_\omega = \bigcup_{n<\omega} L_n\right) \subset \cdots$$

we not only have agreement between L_n and V_n for all $n \leq \omega$, but we can build a model of ZFC in which the continuum hypothesis is true!

This is something we will explore in great detail in this course.

We note that the typical background setting for this course is the entirety of ZFC. However, as and when we need to either get rid of an axiom, or introduce a new one, we will mention it explicitly. We also use the following notational conventions.

Notation. We use **ZFC** to mean we have all the ZFC axioms. As and when we need variations, we do the following.

$$\textbf{ZF} := \text{“ZFC without the Axiom of Choice”}$$
$$\textbf{ZFC} - \textbf{F} := \text{“ZFC without the Axiom of Foundation”}$$
$$\textbf{ZF} - \textbf{F} := \text{“ZF without the Axiom of Foundation”}$$

We also fix some notation for functions.

Notation. For a function f, we write $\text{dom}(f)$ for its domain.

We use one further convention throughout.

Convention. When we write sup, we mean “union over”. Just like in Lean.

We will begin by reviewing the ZFC axioms.

1.1 The Zermelo-Fraenkel Axioms and the Axiom of Choice

We begin by reviewing the Zermelo-Fraenkel Axioms of Set Theory. We work in a first-order language with only one relation symbol, denoted $\in$. In principle, we would want to distinguish between the symbol in the language and its interpretation in any model, but we will not do this in practice. In situations where we really want to be pedantic, we will denote the symbol $\dot{\in}$ and the interpretation $\in$.

We begin with the Axiom of Extensionality.

1.1.1 Extensionality

ZFC Axiom 1 (The Axiom of Extensionality).

$$\forall A \forall B \, [(\forall x \, (x \in A \leftrightarrow x \in B)) \rightarrow (A = B)]$$

We define the $\subseteq$ symbol, used infix as $A \subseteq B$, to be shorthand for the formula $\forall x \ (x \in A \rightarrow x \in B)$. This is not a symbol in the language of set theory; it is merely a convenient abbreviation that we will use hereafter without further discussion.

ZFC Axiom 1 essentially tells us that

$$\forall A \forall B \, ((A \subseteq B \wedge B \subseteq A) \rightarrow A = B)$$

(This is provably equivalent to ZFC Axiom 1.)

We now define the Axiom Scheme of Comprehension.

1.1.2 Comprehension

First, some notation.

Notation. We will denote finite tuples using overlines, so that $\bar{y}$ refers to $(y_0, \ldots, y_{n-1})$. We will typically use bars to bundle together variables.

ZFC is an infinite theory. We call the following an Axiom "Scheme" to underscore the fact that it

is really an infinite family of axioms bundled together.

ZFC Axiom 2 (The Axiom (Scheme) of Comprehension). Let $x, \bar{y}$ be free variables. For all formulae $\varphi(x, \bar{y})$, the following is an axiom:

$$\forall A \forall \bar{y} \exists B \forall x \left[x \in B \leftrightarrow (x \in A \wedge \varphi(x, \bar{y})) \right]$$

Intuitively, this means we can define

$$B = \{x \in A \mid \varphi(x, \bar{y})\}$$

This is an axiom scheme because we can increase the arity of φ: we don't know how many variables y_i are bundled together in $\bar{y}$, only that for any given statement of the axiom, there are finitely many y_i.

Note that ZFC Axiom 1 tells us that we can replace the existential quantifier for B in ZFC Axiom 2 with an existence and uniqueness quantifier without changing its meaning.

Before proceeding further, we define the empty set. In first order logic, structures are required to have non-empty universes. Thus, for every structure (A, E) in the language of set theory, with $E \subseteq (A \times A)$ representing membership, we must have

$$(A, E) \vDash \exists x \, (x = x)$$

The Completeness Theorem then tells us that

$$\vdash \exists x \, (x = x)$$

ie, that the formula $\exists x \, (x = x)$ *must be a theorem* in the language of sets. Hence, ZFC Axiom 1 and ZFC Axiom 2 tell us that

$$\vdash \exists! A \forall x \ (x \notin A)$$

In other words, it is a theorem in the language of sets that there is a unique set that contains no members whatsoever. This unique set is denoted $\emptyset$, and is called the **empty set**. Its existence (and uniqueness) is not a distinct axiom, but a direct consequences of ZFC Axiom 1 and ZFC Axiom 2.

Next, we give the axiom of pairing.

1.1.3 Pairing

ZFC Axiom 3 (The Axiom of Pairing).

$$\forall x \forall y \exists A \, (x \in A \wedge y \in A)$$

ZFC Axiom 3 essentially gives us a way of constructing, for any x and y, the unique set $A = \{x, y\}$ with the property that

$$\forall z \, (z \in A \leftrightarrow (z = x) \vee (z = y))$$

In particular, it allows us to construct, for any x, the set $\{x, x\}$, which, by ZFC Axiom 1, is exactly $\{x\}$. In other words, it tells us that we can stick any x into a set that only contains x.

This allows us to construct pairwise unions, but not arbitrary unions.

1.1.4 Unions

Now, we state the axiom of unions.

ZFC Axiom 4 (The Axiom of Unions). If we denote by $\mathcal{F}$ a family of sets,

$$\forall \mathcal{F} \exists B \forall A \, [A \in \mathcal{F} \rightarrow A \subseteq B]$$

As usual, we can use ZFC Axiom 1, ZFC Axiom 2 to show that

$$\bigcup \mathcal{F} = \{x \mid \exists A \in \mathcal{F} \, (x \in A)\}$$

is well (and uniquely) defined.

Next comes the power set axiom.

1.1.5 Power Sets

ZFC Axiom 5 (The Axiom of Power Sets).

$$\forall A \exists \mathcal{F} \forall X \, [X \subseteq A \to X \in \mathcal{F}]$$

Again, ZFC Axioms 1 and 2 tell us that this is equivalent to

$$\forall A \exists! \mathcal{F} \forall X \, (X \in \mathcal{F} \leftrightarrow X \subseteq A)$$

This justifies the definition

$$\mathcal{P}(A) := \{X \mid X \subseteq A\}$$

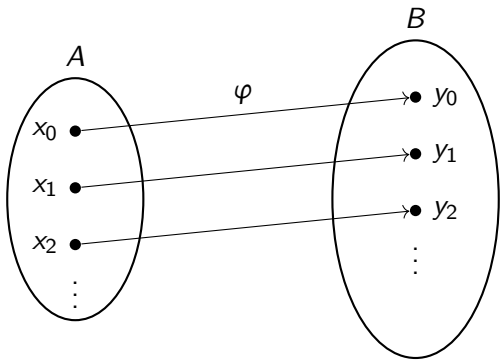
Next comes the second (and last) axiom scheme in ZFC, the axiom scheme of replacement.

1.1.6 Replacement

ZFC Axiom 6 (The Axiom (Scheme) of Replacement). For each formula $\varphi(x, y, \bar{z})$ that does not contain B as a free variable, the following is an axiom:

$$\forall A \forall \bar{z} \, [[\forall x \in A \exists! y \, \varphi(x, y, \bar{z})] \to [[\exists B \forall x \in A \exists y \in B \, \varphi(x, y, \bar{z})]]]$$

Each $x \in A$ has exactly one witness, and what the axiom buys us is that all of those witnesses fit inside a single set B.



B is drawn with room to spare because the axiom only asks for *some* set containing the witnesses; cutting it down to exactly them is what ZFC Axiom 2 does.

We can unpack this slightly complicated formula using ZFC Axiom 1 and ZFC Axiom 2: it essentially tells us that we can define

$$\{y \mid \exists x \in A\ (\varphi(x, y, \bar{z}))\}$$

Ie, if for every x, there is a unique witness y for which the formula $\varphi(x, y, \bar{z})$ holds, we can collect all these witnesses into a set that is contained B. In particular, "the number of such witnesses is *small enough* that it *can be collected into a set*."

Consider the case where $\varphi(x, y, \bar{z})$ defines a function with domain A. If we can define $A \times B = \{(x, y) \mid x \in A \text{ and } y \in B\}$ (which we have yet to show we can do, but let's say we can), then we can put

$$f = \{(x, y) \in A \times B \mid \varphi(x, y, \bar{z})\}$$

which is a set by ZFC Axiom 2. Effectively, ZFC Axiom 6 tells us that *if the domain of a function is a set, then so is its range*. ZFC Axiom 6 is an essential step in the process of actually *defining* functions (or showing that this is something we can *do*).

One situation in which ZFC Axiom 6 is used (and will be used once sufficiently many axioms are stated) is in the situation we talked about at the start of Chapter 1. We'll see that there is a formula $\varphi(n, S)$ so that $\varphi(n, S) \leftrightarrow (n \in \omega \wedge S = V_n)$. Then, ZFC Axiom 6 would imply that

$$\langle V_n \mid n < \omega \rangle$$

really is a sequence. This allows us to define $V_\omega = \bigcup_{n<\omega} V_n$ and continue.

1.1.7 Foundation

ZFC Axiom 7 (The Axiom of Foundation).

$$\forall A\,(A \neq \emptyset \rightarrow \exists x \in A \forall y \in A\, y \notin x)$$

We now state the most interesting of the ZFC axioms: the Axiom of Choice.

1.1.8 Choice

ZFC Axiom 8 (The Axiom of Choice - (AC)).

$$\forall \mathcal{F}\, \exists c\, [\forall A \in \mathcal{F}\ (A \neq \emptyset \to A \in \mathrm{dom}(c) \wedge c(A) \in A)]$$

where $\mathcal{F}$ is a family of sets, c is a function, and $\mathrm{dom}(c)$ is the domain of c. We call c a **Choice Function on** $\mathcal{F}$.

The reason we state the Axiom of Choice at this point in our discussion is that modulo the ZFC Axioms that we have seen so far, the Axiom of Choice is equivalent to each of the following.

1. $\forall A \exists R \subseteq A \times A\ (A, R)$ is a well-ordering.
2. $\forall A \exists \alpha \exists f : A \xrightarrow{\sim} \alpha$, where α is an ordinal.

Note that none of the above axioms imply the existence of the natural numbers, without which virtually none of the mathematics we know and love would be possible. We therefore come to the axiom of infinity, which posits the existence of infinite sets, the 'smallest' of which is the natural numbers.

1.1.9 Infinity

ZFC Axiom 9 (The Axiom of Infinity).

$$\exists I\ (\emptyset \in I \wedge \forall x \in I\ (x \cup \{x\} \in I))$$

A set I as above is called an **inductive set**, and it is an axiom of ZFC that such a set exists.

It is possible to prove that given the existence of inductive sets (as posited above), there is a unique inductive set ω such that for all inductive sets J, $\omega \subseteq J$. Indeed, for any inductive set K, ω can be defined as the intersection of all inductive subsets $J \subseteq K$. It can then be shown that this intersection ω lives in all inductive sets. This minimal inductive set ω is known as the **set of natural numbers**. It is easy to see that the famed axioms of Peano are satisfied by ω.

Intuitively, we'd want to say something like

$$\omega = \bigcap \{k \mid k \text{ is inductive}\}$$

The only problem is, a priori, the above intersection is not a set! This is why we need this axiom.

1.2 Transitive Sets

This is a quick recap section on transitive sets and ordinals. We come to cardinals in Section 1.5.

Definition 1.2.1 (Transitive Set). A set A is **transitive** if $\forall y \in A\ (y \subseteq A)$, ie, if $\forall y \in A \forall x \in y\ (x \in A)$

1.2.1 Ordinals

Definition 1.2.2 (Ordinal). A set α is an **ordinal** iff α is a transitive set and $(\alpha, \in)$ is a well-ordering.

Note that writing $(A, \in)$ really means the structure (A, R) where $R = \{(x, y) \in A \times A \mid x \in y\}$. It's slightly problematic to call $\in$ itself a relation because it isn't a set: it's (strictly speaking) just a symbol, which is interpreted as a set (a relation) in a model of set theory. We have to be careful with these sorts of things.

Example 1.2.3 (Examples of Ordinals).

1. $0 = \emptyset$, $1 = \{0\}$, $2 = \{0, 1\}$, $3 = \{0, 1, 2\}$, etc.
2. $\omega = \{0, 1, 2, \ldots\}$
3. $\omega + 1 = \omega \cup \{\omega\}$, $\omega + 2$, etc.
4. $\omega + \omega$

and so on.

1.2.2 Mostowski Collapse for Well-Orderings

We will eventually prove this in more generality (Theorem 2.2.4), but here's a useful theorem.

Theorem 1.2.4 (Mostowski Collapse Theorem for Well-Orderings). *For all well-orderings (A, R), there is a unique ordinal α such that there is a unique isomorphism π from (A, R) to $(\alpha, \in)$ given by*

$$\pi(y) = \{\pi(x) \mid xRy\}$$

This is a special case of what's called (surprise surprise) the Mostowski Collapse Theorem. Some things we can do include removing the well/linear requirement on the ordering and replace "ordinal" with just "transitive set".

What the theorem above says is that the ordinals form a complete set of representatives for the class of all well-orderings up to isomorphism.

1.3 The V-Hierarchy

Definition 1.3.1 (The V-Hierarchy). Define $V_0 = \emptyset$, $V_{\alpha+1} = \mathcal{P}(V_\alpha)$ for any ordinal α, and $V_\beta = \bigcup_{\alpha<\beta} V_\alpha$ for any limit ordinal β.

One can show that $\alpha \leq \beta \to V_\alpha \subseteq V_\beta$.

We can ask ourselves what exactly ZFC is describing. The answer is, **the V-hierarchy**.

Interestingly,

$$V_\omega \vDash \textbf{ZFC} - \text{infinity} + \neg\text{infinity}$$

Even more interestingly, ZFC doesn't prove its own consistency (if it could, it would be inconsistent, by Gödel), but it comes exactly one axiom away: it can prove the existence of models of everything except for the axiom of infinity.

1.3.1 The Existence of V_ω

It is worth asking ourselves the following question: *Why does V_ω exist?* Or, more precisely, *why is V_ω a set?*

By induction, we know that each V_n is a set. We want to essentially see that the map $n \mapsto V_n$ is a *set* (ie, a *function*). Then, if we take the union of the range of this function, whose domain is the set ω whose existence is given by ZFC Axiom 9, we can define V_ω to be that union.

We invoke ZFC Axiom 6, the Axiom of Replacement. Let $\varphi(x, y)$ be the formula capturing the following facts:

1. y is a function with domain $\text{dom}(y) = x \cup \{x\}$.
2. If $\emptyset \in \text{dom}(y)$, then $y(\emptyset) = \emptyset$.
3. If $z \cup \{z\} \in \text{dom}(y)$, then $z \in \text{dom}(y)$ and $y(z \cup \{z\}) = \mathcal{P}(y(z))$.

The idea is that $y = \langle V_0, V_1, V_2, \ldots, V_n \rangle$[1] for some $n \in \omega$. Then,

$$\textbf{ZFC} \vdash \forall n \in \omega \; \exists! y \; \varphi(n, y)$$

Applying the Axiom of Replacement (ZFC Axiom 6), we know that the map that takes any $n \in \omega$ to its unique witness y for *n is, indeed, a function*. In other words,

$$\langle \langle V_m \mid m \leq n \rangle \mid n < \omega \rangle$$

is a set. From this, we can read off

$$\langle V_n \mid n < \omega \rangle$$

by taking the union of its range. Each $\langle V_m \mid m \leq n \rangle$ is a function with domain $n + 1$, ie, a set of ordered pairs, and each one is contained in the next, so the union of all of them is a function too: the one with domain $\bigcup_{n<\omega} (n + 1) = \omega$ that sends n to V_n.

Remark. Recall that $(V_\omega, \in)$ is shorthand for (V_ω, R), where $R = \{(x, y) \in V_\omega \times V_\omega \mid x \in y\}$. It is possible to show not only that

$$(V_\omega, \in) \vDash \text{ZFC without the Axiom of Infinity}$$

[1] By this, we mean it is a *finite sequence*, ie, a function from $n+1$ to something that takes 0 to V_0, 1 to V_1, and so on up to n.

but the stronger fact that

$$(V_\omega, \in) \vDash \text{ZFC without the Axiom of Infinity but with its } \textit{negation}$$

Thus,

$$\mathbf{ZFC} \vdash \text{ZFC without Infinity but with } \neg\text{Infinity is consistent}$$

Ie, from ZFC, we can deduce that the entirety of ZFC, except with the negation of the axiom of infinity instead of the axiom itself, is consistent.

Here, we end our discussion on the V-hierarchy, and proceed to consider a broader type of object than sets, known as classes.

1.4 Classes

Recall Russell's Paradox: there is no set of all sets. One reason for this (which relies on the Axiom of Foundation) is that if V is the set of all sets, then $V \in V$, so $\{V\}$ has no $\in$-minimal member, which contradicts the Axiom of Foundation. Without the Axiom of Foundation, if V is the set of all sets, then if we set

$$W = \{x \in V \mid x \notin x\}$$

(which is a set by the Axiom of Comprehension), then if $W \in W$, then $W \notin W$, and if $W \notin W$, then $W \in W$, which is a contradiction, meaning there can be no set of all sets.

Nevertheless, we really do want to write things like

$$V = \{x \mid x \text{ is a set}\} = \{x \mid x = x\}$$

We would also like to write things like

$$\mathbf{OR} = \{\alpha \mid \alpha \text{ is an ordinal}\} \qquad \text{and} \qquad \{V_\alpha \mid \alpha \in \mathbf{OR}\}$$

The only problem is, none of these are sets. Indeed, the only guarantee of the Axiom of Comprehension is that

$$\{x \in A \mid \varphi(x, \bar{y})\}$$

is a set; there is no guarantee whatsoever that

$$\{x \mid \varphi(x, \bar{y})\}$$

is a set. It sometimes is, but it isn't always.

There are two ways you can prove that $V = \{x \mid x = x\}$ is not a set. The first is that if it were, then $V \in V$ would hold, meaning $\{V\}$ would have no $\in$-minimal element, contradicting foundation. The second is that if it were, $W = \{x \in V \mid x \notin x\}$ would be a valid set, by comprehension. Then, $W \in W \leftrightarrow W \notin W$, a contradiction.

There are two perspectives we can take. The first is the **external perspective**: this is the perspective from which you would look at models of ZFC. There is also the internal perspective, where you are already inside a model of ZFC. Now this internal perspective also has a notion of logic and set theory inside of it. So there's formalisations of set theory inside models of set theory, and these formalisations have their own notions of models of set theory, and so it goes.

At some point, we will study these things in detail.

For the time being, we will use the term **class** to describe something we might want to express as a set, even if it cannot actually be a set for cardinality reasons.

Note that we would want sets to be classes. Indeed, anything of the form $\{x \mid \varphi(x)\}$, with $\varphi(x)$ a formula in the language of set theory, should be a class. Unfortunately, this is not generous enough: we want, for each set A, that

$$A = \{x \mid x \in A\} = \{x \mid \varphi(x, A)\}$$

to be included as well. In other words, we want to have formulae φ that not only allow us to substitute free variables with input sets x but also allow us to substitute more free variables with parameter sets A.

Formulae and parameters determine the class, but the converse is not true. In other words, if

$$\forall x\,(\varphi(x) \leftrightarrow \varphi'(x))$$

then

$$\{x \mid \varphi(x)\} = \{x \mid \varphi'(x)\}$$

For example, we know that

$$\varphi(x) \leftrightarrow (\varphi(x) \wedge x = x)$$

So you can define the same class using more than one formula.

1.4.1 The Class of Ordinals

Definition 1.4.1 (The Class of Ordinals). Define

$$\mathbf{OR} := \{\alpha \mid \alpha \text{ is an ordinal}\}$$

We can show that **OR** is a class.

We can then say that the 'mapping'

$$\alpha \mapsto V_\alpha$$

is a **class function**: it takes elements of a class and maps them to elements of another class.

We can represent this mapping as

$$\{(\alpha, V_\alpha) \mid \alpha \in \mathbf{OR}\}$$

Now we can ask ourselves the following question: for which formulae φ is this set equal to $\{x \mid \varphi(x)\}$?

Let's give two such formulae based on the following.

Theorem 1.4.2. *$\forall \gamma \in \mathbf{OR}$, $\forall S \in V$, TFAE:*

1. $S = V_\gamma$
2. *$\exists g$ a function with the properties that*
   (a) *The domain of g is $\gamma + 1$*
   (b) $g(0) = \emptyset$

(c) $\forall \alpha < \gamma$, $g(\alpha + 1) = \mathcal{P}(g(\alpha))$

(d) $\forall \beta \leq \gamma$ *with* β *a limit ordinal,* $g(\beta) = \bigcup_{\alpha<\beta} g(\alpha)$

(e) $g(\gamma) = S$

3. *Every function* g *exhibiting properties (a)-(d) above also exhibits property (e).*

1.4.2 Well-Foundedness

We now define the class of well-founded sets.

Definition 1.4.3 (The Class of Well-Founded Sets). Define

$$\mathbf{WF} := \bigcup_{\alpha \in \mathbf{OR}} V_\alpha$$

That is, **WF** consists of all x such that there is some ordinal α with $x \in V_\alpha$.

WF is a class.

We will eventually see that the axiom of foundation is equivalent to saying that $V = \mathbf{WF}$ (under **ZF** without foundation).

Proposition 1.4.4. *For all* $\gamma \in \mathbf{OR}$,

$(1)_\gamma$ $\forall \alpha \leq \gamma$, $V_\alpha \subseteq V_\gamma$

$(2)_\gamma$ V_γ *is a transitive set*

Proof. We argue by induction on ordinals γ.

For the base step ($\gamma = 0$), both $(1)_0$ and $(2)_0$ are clear.

For the successor step, if $\gamma = \beta + 1$,

$(1)_\gamma$ Let $\alpha \leq \beta$ and $x \in V_\alpha$. By $(1)_\beta$, $x \in V_\beta$. By $(2)_\beta$, $x \subseteq V_\beta$. So, $x \in \mathcal{P}(V_\beta) = V_{\beta+1} = V_\gamma$.

$(2)_\gamma$ Let $y \in V_\gamma = \mathcal{P}(V_\beta)$. Then, $y \subseteq V_\beta$. By $(1)_\gamma$, $y \subseteq V_\gamma$.

Finally, for the limit step, ie, if γ is a limit ordinal, then $(1)_\gamma$ and $(2)_\gamma$ both follow from the fact

that $\forall \beta < \gamma$, $(1)_\beta$ and $(2)_\beta$ both hold. (This is because in the limit case, constructing V_γ is just done by taking unions.) □

Definition 1.4.5 (Rank). Define, for $x \in \mathbf{WF}$, the **rank** of x, denoted $\text{rank}(x)$, to be the least $\alpha \in \mathbf{OR}$ such that $x \in V_{\alpha+1}$.

Indeed, this is equivalent to $\text{rank}(x)$ being the least $\alpha \in \mathbf{OR}$ such that $x \subseteq V_\alpha$.

Proposition 1.4.6. *The mapping*

$$x \mapsto \text{rank}(x)$$

is a class surjection from **WF** *to* **OR**.

We won't prove this, but we'll say it's easy enough to prove and leave it at that.

Proposition 1.4.7. $\forall \beta \in \mathbf{OR}$,

$$V_\beta = \{x \in \mathbf{WF} \mid \text{rank}(x) < \beta\}$$

Proof sketch. You can just do induction on β and apply the definition of the rank (Definition 1.4.5). □

It turns out that we can tell whether a set lies in **WF** by looking only at its members.

Proposition 1.4.8. $\forall y\, (y \in \mathbf{WF} \leftrightarrow y \subset \mathbf{WF})$

Proof.

$\rightarrow$ Let $\beta = \text{rank}(y)$. Then $y \subseteq V_\beta \subset \mathbf{WF}$.

$\leftarrow$ Assume $y \subset \mathbf{WF}$. Define $f : y \to \mathbf{OR}$ by $f(x) = \text{rank}(x) + 1$. We're using replacement to know that $f \in V$.

Let $\beta = \sup_{x \in y} f(x)$. Then, $y \subseteq V_\beta$. So $y \in \mathcal{P}(V_\beta) = V_{\beta+1} \subset \mathbf{WF}$.

□

Members have strictly smaller rank than the sets that contain them.

Proposition 1.4.9.

$$\forall y \in \mathbf{WF},\ \forall x \in y,\ [x \in \mathbf{WF} \wedge \mathrm{rank}(x) < \mathrm{rank}(y)]$$

Proof. We know $x \in y \subseteq V_{\mathrm{rank}(y)} \subset \mathbf{WF}$, so $x \in \mathbf{WF}$. This also shows that $\mathrm{rank}(x)+1 \leq \mathrm{rank}(y)$, so $\mathrm{rank}(x) < \mathrm{rank}(y)$. □

From this, we can compute the rank of a set from the ranks of its members.

Proposition 1.4.10 (The Rank Formula).

$$\forall y \in \mathbf{WF},\ \mathrm{rank}(y) = \sup_{x \in y}(\mathrm{rank}(x) + 1)$$

Proof. Put $\rho = \mathrm{rank}(y)$ and $\sigma = \sup_{x \in y}(\mathrm{rank}(x) + 1)$. By Proposition 1.4.9, if $x \in y$, then $\mathrm{rank}(x) < \mathrm{rank}(y) = \rho$, so $\mathrm{rank}(x) + 1 \leq \rho$. So $\sigma \leq \rho$.

It suffices to show that $y \subseteq V_\sigma$ to conclude that $\rho \leq \sigma$. Now $V_\sigma = \{x \mid \mathrm{rank}(x) < \sigma\}$. Every $x \in y$ has $\mathrm{rank}(x) + 1 \leq \sigma$, by the definition of σ as a supremum, and so $\mathrm{rank}(x) < \sigma$. So y is indeed a subset. □

It turns out that the ordinals are their own ranks.

Proposition 1.4.11.

$$\forall \beta \in \mathbf{OR},\ [\beta \in \mathbf{WF} \wedge \mathrm{rank}(\beta) = \beta]$$

Proof. We argue by induction on β. Base case is trivial. In the inductive case, $\beta \subseteq \mathbf{WF}$, so by previous prop, $\beta \in \mathbf{WF}$. So,

$$\mathrm{rank}(\beta) = \sup_{\alpha < \beta}(\mathrm{rank}(\alpha) + 1)$$

$$= \sup_{\alpha<\beta}(\alpha+1)$$
$$= \beta$$

□

It follows that each V_β contains exactly the ordinals below β.

Proposition 1.4.12.

$$\forall \beta \in \mathbf{OR},\ [V_\beta \cap \mathbf{OR} = \beta]$$

Proof. We know

$$\begin{aligned} V_\beta \cap \mathbf{OR} &= \{x \in \mathbf{WF} \mid \mathrm{rank}(x) < \beta\} \cap \mathbf{OR} \\ &= \{\alpha \in \mathbf{OR} \mid \mathrm{rank}(\alpha) < \beta\} \\ &= \{\alpha \in \mathbf{OR} \mid \alpha < \beta\} \\ &= \beta \end{aligned}$$

□

1.4.3 Well-Foundedness of Membership

We now say what it means for a relation on a class to be well-founded. Say A is a class and R is a class binary relation[2]. When we write (A, R), we really mean the class structure

$$(A, R \cap (A \times A))$$

Definition 1.4.13 (Well-Foundedness). We say that (A, R) is **well-founded** if for every nonempty set $S \subseteq A$, $\exists x \in S$ such that $\forall y \in S$, $\neg(y\, R\, x)$.
We say that (A, R) is **ill-founded** if it is not well-founded.

Membership on a well-founded set is, reassuringly, well-founded.

[2] ie, a binary relation on elements of the class A

Proposition 1.4.14. *$\forall A \in \mathbf{WF}$, $(A, \in)$ is well-founded.*

Proof. Let $S \subseteq A$, $S \neq \emptyset$. Denote

$$\alpha = \min_{x \in S} \operatorname{rank}(x)$$

This makes sense: every $x \in S$ is a member of $A \in \mathbf{WF}$, so $x \in \mathbf{WF}$ by Proposition 1.4.9 and has a rank, and **OR** is well-ordered. Pick $x \in S$ with $\operatorname{rank}(x) = \alpha$. If some $y \in S$ had $y \in x$, then $\operatorname{rank}(y) < \operatorname{rank}(x) = \alpha$ by Proposition 1.4.9 again, contradicting the minimality of α. So x is $\in$-minimal in S. □

Before going further, we give a name to the smallest transitive set containing a given set; that there is one is the content of a lemma below.

Definition 1.4.15 (Transitive Closure). Given a set Y, the **transitive closure** of Y, denoted $\operatorname{trcl}(Y)$, is the smallest transitive set T such that $Y \subseteq T$. In particular, $\operatorname{trcl}(\{x\})$ is the smallest transitive set containing x as a member.

The converse of Proposition 1.4.14 fails, and the transitive closure is exactly what the failure is hiding.

Non-Example 1.4.16 (An Ill-Founded Relation). One way Foundation can fail is if we have two sets $x \neq y$ such that $x = \{y\}$ and $y = \{x\}$.

In this case, $(x, \in)$ and $(y, \in)$ are both well-founded (just because x and y are both singletons). But $x, y \notin \mathbf{WF}$, because if they were, then they would *both* be, so they would both have ranks, and since $x \in y$, $\operatorname{rank}(x) < \operatorname{rank}(y)$ and similarly, since $y \in x$, $\operatorname{rank}(y) < \operatorname{rank}(x)$, which is impossible.

In this example, the transitive closure of x is $\{x, y\}$, which is also the transitive closure of y. Moreover, $(\{x, y\}, \in)$ is ill-founded.

For transitive sets, on the other hand, there is nothing to hide.

Proposition 1.4.17. *For every transitive set A, if $(A, \in)$ is well-founded, then $A \in \mathbf{WF}$.*

Proof. Fix A a transitive set and assume $(A, \in)$ is well-founded. To see that $A \in \mathbf{WF}$, it suffices to show that $A \subset \mathbf{WF}$.

Suppose not. Let $S = A \setminus \mathbf{WF}$. Then, S is a nonempty subset of A. As $(A, \in)$ is well-founded, there is some $x \in S$ that is $\in$-minimal in S. Then $x \notin \mathbf{WF}$, so $x \not\subset \mathbf{WF}$.

Pick $y \in x \setminus \mathbf{WF}$. Then, $y \in x \in A$. Since A is transitive, $y \in A$. Thus, $y \in A \setminus \mathbf{WF} = S$, and $y \in x$. This contradicts our choice of x. □

The transitive closure can be built from below, in ω steps.

Lemma 1.4.18. *Given a set Y, put*

$$Z_0 = Y$$
$$Z_{n+1} = \bigcup Z_n$$
$$Z_\omega = \bigcup_{n<\omega} Z_n$$

Then, $\mathrm{trcl}(Y) = Z_\omega$.

We can now deliver on the promise made earlier in this section: over $\mathbf{ZF} - \mathbf{F}$, Foundation says precisely that $V = \mathbf{WF}$.

Proposition 1.4.19 (Assuming $\mathbf{ZF} - \mathbf{F}$). *TFAE:*

(1) *Foundation: every nonempty S has an $\in$-minimal member*

(2) *For all A, $(A, \in)$ is well-founded*

(3) $V = \mathbf{WF}$

Proof. (1) $\iff$ (2) is clear: taking $A = S$ in (2) gives (1), and conversely a nonempty $S \subseteq A$ is in particular a nonempty set, so (1) hands us an $\in$-minimal member of S.

(3) $\implies$ (2) follows from Proposition 1.4.14.

Let's now assume (2) and prove (3). $\mathbf{WF} \subseteq V$, so we only need $V \subseteq \mathbf{WF}$. Let $x \in V$ and let $A = \mathrm{trcl}(\{x\})$. By Proposition 1.4.17, since A is transitive and $(A, \in)$ is well-founded, by our

assumption (2), we get that $A \in \mathbf{WF}$. So $x \in A \subset \mathbf{WF}$. Thus, $x \in \mathbf{WF}$. □

1.5 Cardinals and the Axiom of Choice

In this section, we discuss how big the levels of the V-hierarchy are. The answer is elementary at the bottom of the hierarchy, but by the time we reach $V_{\omega+1}$ it already depends on the Axiom of Choice. We end by defining two hierarchies of cardinals, only one of which needs choice, and computing the size of every level of the hierarchy in terms of the second.

1.5.1 The Size of V_ω

We begin with the finite levels.

Proposition 1.5.1. *For all $n < \omega$, the set V_n is finite.*

Indeed, their sizes are 0, 2^0, 2^{2^0}, and so on.

So V_ω is the first infinite level. It turns out to be countable, and there are two ways of seeing it.

Proposition 1.5.2. *V_ω is countable.*

Proof using AC. Write

$$V_\omega = \bigcup_{n<\omega} V_n$$

AC tells us that a countable union of countable sets is countable. □

Proof that avoids AC. By recursion on $n < \omega$, we can define $R_n \subseteq V_n \times V_n$ so that

(1) (V_n, R_n) is a well-ordering

(2) R_{n+1} end-extends R_n

Then, we'll be able to put $R_\omega = \bigcup_{n<\omega} R_n$ and show that R_ω is a well-ordering of type ω.

Since (V_n, R_n) is a well-ordering of a finite set, we may enumerate V_n as

$$a_0 \, R_n \, a_1 \, R_n \, \cdots \, R_n \, a_{k-1}$$

where $k = |V_n|$. Every member of $V_{n+1} = \mathcal{P}(V_n)$ is a subset of V_n, so we may write any $x, y \in V_{n+1}$ as

$$x = \{a_i \mid i \in I\} \qquad \text{and} \qquad y = \{a_i \mid i \in J\}$$

for unique $I, J \subseteq k$. Now, put $x \, R_{n+1} \, y$ iff

(i) $x, y \in V_n$ and $x \, R_n \, y$.

(ii) $x \in V_n$ and $y \in V_{n+1} \setminus V_n$.

(iii) $x, y \in V_{n+1} \setminus V_n$, and $i \in J$, where i is the largest natural number lying in exactly one of I and J.

Case (iii) makes sense because $x \neq y$ forces $I \neq J$, so some natural number lies in exactly one of them and there are only finitely many candidates. The relation it defines is the ordering of the subsets of V_n by binary numerals, reading x as the string whose ith digit records whether $i \in I$, so it is a linear ordering.

So R_{n+1} is R_n followed by a linear ordering of the finitely many sets in $V_{n+1} \setminus V_n$, which makes it a well-ordering and gives us (1). It agrees with R_n on V_n and places all of V_n before all of $V_{n+1} \setminus V_n$, so V_n is an initial segment of (V_{n+1}, R_{n+1}), which gives us (2).

The point of the construction is that R_{n+1} is defined outright from R_n, with no choices made along the way, and the recursion starts from $R_0 = \emptyset$ on $V_0 = \emptyset$. The R_n increase, so R_ω linearly orders V_ω. Chaining (2) makes V_n an initial segment of (V_m, R_m) for every $m \geq n$, so each $x \in V_\omega$ has all of its R_ω-predecessors inside the finite V_n it came from, and every proper initial segment of R_ω is therefore finite. Since $\omega \subseteq V_\omega$, the ordering is infinite, so sending each x to its number of predecessors is an isomorphism onto ω. $\square$

1.5.2 Cardinality Without Choice

The reason the second proof is worth having is that without AC, countable unions are badly behaved.

Remark. If ZF is consistent, then so is ZF + "there's a countable union of pairs that has no cardinality".

The same problem arises one level further up.

Remark. Without AC, we don't know whether the set $V_{\omega+1} = \mathcal{P}(V_\omega)$ even has a *cardinality*. With AC,

$$|V_{\omega+1}| = |\mathcal{P}(V_\omega)| = 2^{\aleph_0}$$

where $\aleph_0 = \omega = |V_\omega|$.

1.5.3 Cardinal Successors

It turns out that the successor of a cardinal can still be defined without choice, provided we go about it the right way.

Given AC, we know that $2^\kappa > \kappa$ and we can define κ^+ to be the least cardinal strictly greater than κ. Without AC, we still know, given an infinite cardinal κ, that

$$\{R \subseteq \kappa \times \kappa \mid (\kappa, R) \text{ is a well-ordering}\}$$

is a set. We can still have Mostowski Collapse for well-orderings. We can then conclude that

$$\{\text{types}(\kappa, R) \mid (\kappa, R) \text{ is a well-ordering}\}$$

is a set of ordinals. Then, we can set κ^+ to simply be the supremum of this set.

1.5.4 The Alephs and the Beths

We can now define the two standard hierarchies of cardinals, of which only the second needs the Axiom of Choice.

Without AC, define

$$\begin{aligned}\aleph_0 &:= \omega \\ \aleph_{\alpha+1} &:= (\aleph_\alpha)^+ \\ \aleph_\beta &:= \sup_{\alpha<\beta}(\aleph_\alpha) \text{ if } \beta \text{ is a limit ordinal}\end{aligned}$$

With AC, define

$$\begin{aligned}\beth_0 &:= \omega \\ \beth_{\alpha+1} &:= 2^{\beth_\alpha} \\ \beth_\beta &:= \sup_{\alpha<\beta}(\beth_\alpha) \text{ if } \beta \text{ is a limit ordinal}\end{aligned}$$

It's clear `sorry`

With the $\beth$s in hand, we can say exactly how big every level of the hierarchy is, from V_ω onwards.

Proposition 1.5.3.

$$\forall \eta \in \mathbf{OR},\ |V_{\omega+\eta}| = \beth_\eta$$

Proof using AC. Do induction on η. For the base case, $|V_\omega| = \aleph_0 = \beth_0$, as V_ω is countable.

For the successor step, if $|V_{\omega+\eta}| = \beth_\eta$, then

$$|V_{\omega+\eta+1}| = |\mathcal{P}(V_{\omega+\eta})| = 2^{|V_{\omega+\eta}|} = 2^{\beth_\eta} = \beth_{\eta+1}$$

Finally, let η be a limit ordinal. Then $\omega+\eta$ is a limit ordinal too, and every ordinal below it is either below ω or of the form $\omega+\alpha$ for some $\alpha<\eta$, so

$$V_{\omega+\eta} = \bigcup_{\alpha<\eta} V_{\omega+\alpha}$$

Each $V_{\omega+\alpha}$ is a subset of $V_{\omega+\eta}$, so $\beth_\alpha \leq |V_{\omega+\eta}|$ for every $\alpha<\eta$, and hence $\beth_\eta = \sup_{\alpha<\eta}\beth_\alpha \leq |V_{\omega+\eta}|$. Conversely, $V_{\omega+\eta}$ is a union of $|\eta|$ sets, each of cardinality at most $\beth_\eta$, so (using AC) $|V_{\omega+\eta}| \leq |\eta| \cdot \beth_\eta$. It remains to see that $|\eta| \leq \beth_\eta$, and in fact $\alpha \leq \beth_\alpha$ for every ordinal α: this is clear for $\alpha = 0$, it passes through limits by taking suprema, and it passes through successors

because $\beth_{\alpha+1} = 2^{\beth_\alpha} > \beth_\alpha \geq \alpha$ forces $\beth_{\alpha+1} \geq \alpha + 1$. So $|V_{\omega+\eta}| \leq \beth_\eta \cdot \beth_\eta = \beth_\eta$. □

A word of caution before we go on.

Warning. If we continue to naïvely indulge ourselves the idealistic notion that we can use $\aleph$s to denote cardinals and ωs to denote the corresponding ordinals, we will soon encounter a world of confusion and pain.
The two notations will often be used interchangeably, just as they are in practice. We will try our best, but we might not always succeed.

The proposition indexes the levels from V_ω rather than from V_0, but the shift by ω is invisible from ω^2 onwards.

Proposition 1.5.4 (without AC).

$$\forall \eta \geq \omega^2,\ (\omega + \eta = \eta)$$

Note that $\omega^2 = \omega \cdot \omega$; by ordinal arithmetic, it's a countable ordinal.

Proof. First, write $\eta = \omega^2$. Then,

$$\omega + \omega^2 = \omega\,(1 + \omega) = \omega \cdot \omega = \omega^2$$

Now, in general, for $\eta = \omega^2 + \delta$, then $\omega + (\omega^2 + \delta) = (\omega + \omega^2) + \delta = \omega^2 + \delta$. □

Chapter 2

Induction, Recursion and Collapse

In Chapter 1, we saw that membership is a well-founded relation on **WF**, and that Foundation says precisely that this is true everywhere. In this chapter, we study well-founded relations on classes in general, with a view to doing induction and recursion along them. We begin with a second property a class relation may or may not have, which will be needed alongside well-foundedness.

2.1 Induction and Recursion

In what follows, let A be a class and R be a class relation on A. We want to prove things about the members of A by induction along R and define things on them by recursion along R, and this section says what R has to look like for that to work, and then does it.

2.1.1 Set-Like Relations

The property we want is that looking below any one point of A only ever sees a set's worth of A.

Definition 2.1.1 (Set-Like). We say (A, R) is **set-like** iff

$$\forall y \in A, \{x \in A \mid x \, R \, y\}$$

is a set.

The example we care about most has this property for the most basic of reasons.

Example 2.1.2 (Membership). $(A, \in)$ is set-like for every class A.

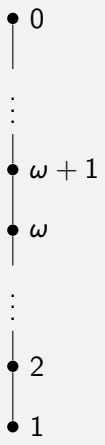
In fact, the fact in the above example is **equivalent** to the axiom scheme of comprehension (ZFC Axiom 2)! Indeed, if $A = \{x \mid \varphi(x, \bar{z})\}$ and $y \in A$, then

$$\{x \in A \mid x \in y\} = \{x \in y \mid \varphi(x, \bar{z})\}$$

So "membership is set-like" is a fancy way of saying "Comprehension".

But not all relations are set-like.

Non-Example 2.1.3. Let $A = \mathbf{OR}$. Define R by the following picture:



That is, it looks a bit like $(\mathbf{OR} \setminus \{0\}, <)$.

2.1.2 Proofs by Induction

We now have another *scheme* of facts. In what follows, start with A a class and $R \subseteq A \times A$ a class binary relation. Assume (A, R) is well-founded and set-like.

The first fact is that well-foundedness, which speaks of subsets of A, extends to subclasses.

Proposition 2.1.4 (Minimal Members of Subclasses). *If S is a nonempty subclass of A, then S has an R-minimal member.*

Proof. We may assume that $R \subseteq A \times A$. Since $S \neq \emptyset$, we have some $z \in S$. Let T be the

R-closure of $\{z\}$. That means that we can define, by recursion on $n < \omega$, the sets

$$T_0 = \{z\} \qquad T_{n+1} = \bigcup_{y \in T_n} \{x \mid x\, R\, y\}$$

The fact that (A, R) is set-like tells us that the union above is a union of sets, and therefore, a set.

Next, put $T = \bigcup_{n<\omega} T_n$. This is the smallest R-closed set that has z as a member.

Now, $S \cap T$ is a nonempty subset of A. So it has an R-minimal member, say x. Here, using the fact that T is R-closed, it is easy to show that x is R-minimal among members of S. Otherwise, we'd have $y\, R\, x$, with $y \in S$. Since $x \in T$ and T is R-closed, we have $y \in T$. This contradicts R-minimality of x in $S \cap T$. □

From this we get induction along R.

Theorem 2.1.5 (Theorem Scheme about Proofs by Induction). *Let $\varphi(u, w)$ be a formula in the language of set theory, and assume z is a set.*
Assume that for all y, if y satisfies the formula that defines A (that is, $\forall y \in A$),

$$[(\forall x\, R\, y,\ \varphi(x, z)) \to \varphi(y, z)]$$

We can then conclude that $\forall x \in A,\ \varphi(x, z)$.

Proof. Deny for contradiction. Let $S = \{x \in A \mid \neg\varphi(x, z)\}$. By the previous proposition, as S is a nonempty subclass of A, S has an R-minimal member, say y. Then, $\forall x\, R\, y,\ \varphi(x, z)$.

By assumption, $\varphi(y, z)$, ie, $y \notin S$. This is a contradiction. □

2.1.3 Definitions by Recursion

Notice that in mathematics, we define things by recursion and prove things by induction. But when we explain how everything works, we usually go backwards!

Theorem 2.1.6 (Theorem Scheme about Definitions by Recursion). *Let $F : A \times V \to V$ be a class function. Then, there is a class function $G : A \to V$ such that $\forall y \in A$,*

$$G(y) = F(y, G \restriction \{x \mid x \, R \, y\})$$

Moreover, G is unique in the sense that if $H : A \to V$ is any class function such that

$$\forall y \in A,\ H(y) = F(y, H \restriction \{x \mid x \, R \, y\})$$

then $\forall x \in A,\ H(x) = G(x)$.

Warning. Important subtlety: the statement "there exists a class function" is not a valid statement in the language of set theory, as we cannot quantify over *classes*. What *is* valid, though, is saying something like "such and such a G satisfies such and such a property", where the definition shall come from the proof below.

The proof builds G out of set-sized pieces, which deserve a name.

Definition 2.1.7 (Approximation). We call g an **approximation** iff g is a set function whose domain is an R-closed subset of A (ie, if $y \in \mathrm{dom}(g)$ and $x \, R \, y$, then $x \in \mathrm{dom}(g)$) and $\forall y \in \mathrm{dom}(g)$, $g(y) = F(y, g \restriction \{x \mid x \, R \, y\})$.

Proof. We'll prove uniqueness and then existence.

Uniqueness. Suppose g and h are approximations. Then, $\forall y \in \mathrm{dom}(g) \cap \mathrm{dom}(h)$, $g(y) = h(y)$.

You can prove this using R-induction.

Existence. We'll show that for all $z \in A$, there is some approximation g_z with $\mathrm{dom}(g_z)$ being the R-closure of $\{z\}$. These g_z, if we can show them to exist, are unique (by above).

Use R-induction. The induction hypothesis says that we have g_y as required for each $y \, R \, z$. Let $h = \bigcup_{y \, R \, z} g_y$. We can see that $\mathrm{dom}(h)$ is the R-closure of $\{y \mid y \, R \, z\}$.

Put $g_z = h \cup \{(z, F(z, h \restriction \{x \mid x \, R \, z\}))\}$.

This works! Then, put $G = \bigcup_{z \in A} g_z$.

Indeed, h is a set, by Replacement, since $y \mapsto g_y$ is a class function on the set $\{y \mid y\, R\, z\}$; it is a function, because any two of the g_y agree on their common domain by uniqueness; and it is an approximation, because a union of R-closed sets is R-closed and each g_y satisfies the defining equation on its own domain. Moreover, $z \notin \mathrm{dom}(h)$, as otherwise z would lie on an R-cycle, which well-foundedness forbids. So g_z is a function, its domain $\mathrm{dom}(h) \cup \{z\}$ is the R-closure of $\{z\}$, and it satisfies the defining equation at z by construction and below z because h does.

Finally, G is a class function because the g_z agree wherever their domains overlap, again by uniqueness, and every $y \in A$ lies in $\mathrm{dom}(g_y)$, so $G(y) = g_y(y) = F(y, g_y \restriction \{x \mid x\, R\, y\}) = F(y, G \restriction \{x \mid x\, R\, y\})$, since G agrees with g_y on $\mathrm{dom}(g_y) \supseteq \{x \mid x\, R\, y\}$. Uniqueness of G in the sense of the theorem is R-induction once more: if H satisfies the same equation, then $H(x) = G(x)$ for all $x \in A$.

□

The proof also tells us how to compute G, which is what makes "$G(x) = s$" a statement in the language of set theory after all.

Lemma 2.1.8. *Given any $x \in A$ and set s, TFAE:*

1. $G(x) = s$
2. *There exists an approximation g with $x \in \mathrm{dom}(g)$ and $g(x) = s$.*
3. *For all approximations g, if $x \in \mathrm{dom}(g)$ then $g(x) = s$.*

2.2 Mostowski Collapse

In Theorem 1.2.4 we saw that every well-ordering is isomorphic to a unique ordinal, by a map that sends each point to the set of images of its predecessors. In this section we prove the general version promised there: the ordering hypothesis is dropped, and "ordinal" becomes "transitive class". One more property of the relation is needed to make the map injective, and we begin with it.

2.2.1 Extensional Relations

Definition 2.2.1 (Extensionality). (A, R) is **extensional** iff $\forall y, z \in A$,

$$\{x \in A \mid x \, R \, y\} = \{x \in A \mid x \, R \, z\} \to y = z$$

This is just like the axiom of extensionality, but worded as a property of sets and relations.

Not all relations are extensional.

Non-Example 2.2.2 (Non-Extensional Relations). If $A = \{\{1\}, \{2\}\}$, then $(A, \in)$ is not extensional. The reason is that

$$\{x \in A \mid x \in \{1\}\} = \emptyset = \{x \in A \mid x \in \{2\}\}$$

but obviously $\{1\} \neq \{2\}$.

But many are.

Example 2.2.3 (Extensional Relations).

1. If M is a transitive class, then $(M, \in)$ is extensional. Indeed, suppose $y, z \in M$ and $y \neq z$. Since M is transitive, $y, z \subseteq M$, so $y = y \cap M$ and $z = z \cap M$. Thus

   $$y \cap M = \{x \in M \mid x \in y\} \neq \{x \in M \mid x \in z\} = z \cap M$$

2. If (A, R) is a class linear ordering (ie, a linear ordering on the class A), then (A, R) is extensional.
   For instance, if $A \subseteq \mathbf{OR}$ and R is the usual ordering of ordinals (write R as $\in$ or $<$), then (A, R) is extensional.

2.2.2 The Collapse Theorem

We are now ready to state and prove Mostowski Collapse. Note that it will use neither Choice nor Foundation.

Theorem 2.2.4 (Mostowski Collapse Theorem Scheme). *Let (A, R) be a class structure that's well-founded, set-like and extensional. Define, using the theorem scheme on definitions by recursion, the mapping $G : A \to V$ by*

$$\forall y \in A, G(y) = \{G(x) \mid x \in A \text{ and } x\, R\, y\}$$

Write the range of G as M. Then,

1. *M is a transitive class*
2. *G is a class isomorphism from (A, R) to $(M, \in)$*
3. *If M' is a transitive class and $G' : (A, R) \simeq (M', \in)$, then $M' = M$ and $G' = G$.*

*Note that G is referred to as the **Mostowski Collapse Isomorphism**.*

Remark. When we say that $M' = M$ and $G' = G$ above, we are not saying that their defining formulae are the same. In other words, their equality is not definitional.

Proof. We may assume $R \subseteq A \times A$.

1. Say $a \in b \in M$. We must see $a \in M$. Then, $b \in M$. As M is the range of G, there's some $y \in A$ such that $b = G(y)$. As $G(y) = \{G(x) \mid x\, R\, y\}$, there's some $x\, R\, y$ such that $a = G(x)$. Now $a = G(x) \in M$.

2. We need to show G is an order-preserving bijection.

   Bijectivity. It's clear that G is surjective, because M is the range of G. To show G is injective, it is enough to show that for every $x \in A$,

   $$\forall x' \in A\, [G(x) = G(x') \to x = x'] \tag{2.2.1}$$

   Let's prove (2.2.1) by induction on R. Let $y \in A$ and assume (2.2.1) holds for all $x\, R\, y$. We'll now prove that (2.2.1) holds for y as well.

   Fix any $y' \in A$ and assume that $G(y) = G(y')$. We need to prove $y = y'$. Because (A, R) is extensional, it is enough to show that

   $$\{x \mid x\, R\, y\} = \{x' \mid x'\, R\, y'\}$$

($\subseteq$) Suppose $x\,R\,y$. Then

$$G(x) \in G(y) = G(y') = \{G(x') \mid x'\,R\,y'\}$$

So there's some $x'\,R\,y'$ with $G(x) = G(x')$. By (2.2.1) for x, we know that $x = x'$. Thus $x\,R\,y'$.

($\supseteq$) Suppose $x'\,R\,y'$. Then,

$$G(x') \in G(y') = G(y) = \{G(x) \mid x\,R\,y\}$$

So there's some $x\,R\,y$ such that $G(x') = G(x)$. By (2.2.1) for x, $x'\,R\,y$.

Monotonicity. Suppose $G(x) \in G(y)$. By definition of $G(y)$, there is some $x'\,R\,y$ such that $G(x) = G(x')$. As G is injective, we know that $x = x'$. Thus, $x\,R\,y$.

3. Suppose M' is a transitive class and $G' : (A, R) \simeq (M', \in)$. We need $M = M'$ and $G = G'$.

   Since M is the range of G and M' is the range of G', it is enough to show that $G = G'$. By the uniqueness of recursive definitions (Theorem 2.1.6), it is enough to see that G' satisfies the same defining equation as G. That is, $G'(y) = \{G'(x) \mid x\,R\,y\}$.

   ($\supseteq$) $x\,R\,y \to G'(x) \in G'(y)$ just because G' preserves order (by assumption).

   ($\subseteq$) Suppose $a \in G'(y)$. Now $G'(y) \in M'$ trivially because M' is the range of G'. Since M' is a transitive class, $G'(y) \subseteq M'$. Hence, $a \in M'$. So we have some $x \in A$ such that $a = G'(x)$. Now $G'(x) \in G'(y)$. Since G' is an isomorphism, this implies $x\,R\,y$.

□

2.2.3 Examples

Let's study some examples. The first collapses a proper class, and what it collapses onto is a class we have had since Chapter 1.

Example 2.2.5 (Sequences of Ordinals). Consider the following classes:

$$^{<\omega}\mathbf{OR} = \{f \mid \exists n < \omega,\ f : n \to \mathbf{OR}\}$$

$$\mathbf{OR}^{<\omega} = \{t \mid \exists n < \omega,\ t \text{ is an } n\text{-tuple of ordinals}\}$$

$$[\mathbf{OR}]^{<\omega} = \{s \mid \exists n < \omega, |s| = n \text{ and } s \subset \mathbf{OR}\}$$

The above are pairwise in bijection. No parameters are necessary for anything here.

Let's consider $[\mathbf{OR}]^{<\omega}$. Say $s \in [\mathbf{OR}]^{<\omega}$. Say $n < \omega$ and $s \in [\mathbf{OR}]^n$. List s in decreasing order as $s_0 > s_1 > \cdots > s_{n-1}$, where $s = \emptyset$ if $n = 0$. Identify s with the n-tuple $(s_0, \ldots, s_{n-1})$. Let $<_{\text{lex}}$ denote the lexicographic ordering on members of $[\mathbf{OR}]^{<\omega}$ under this identification, where a tuple counts as smaller than any tuple properly extending it. We see that

- $<_{\text{lex}}$ is a class well-ordering of $[\mathbf{OR}]^{<\omega}$ that is definable without parameters
- $\left([\mathbf{OR}]^{<\omega}, <_{\text{lex}}\right)$ is well-founded and extensional
- $<_{\text{lex}}$ is set-like: given $s \in [\mathbf{OR}]^{<\omega}$,

$$\left\{t \in [\mathbf{OR}]^{<\omega} \;\middle|\; t <_{\text{lex}} s\right\} \subseteq [\max(s) + 1]^{<\omega}$$

We can apply the Mostowski Collapse Theorem Scheme (Theorem 2.2.4) to get a map

$$G : \left([\mathbf{OR}]^{<\omega}, <_{\text{lex}}\right) \simeq (\mathbf{OR}, <)$$

In particular, G is a class bijection between $[\mathbf{OR}]^{<\omega}$ and $\mathbf{OR}$ that is definable without parameters.

The second example is less a computation than a technique, one we will keep coming back to.

Example 2.2.6 (ZFC). Consider any $\alpha \in \mathbf{OR}$. Recall that V_α is a transitive set, so $(V_\alpha, \in)$ is extensional, set-like and well-founded.

Now, you can formalize all of mathematics in ZFC, including Model Theory—in particular, the downwards Löwenheim-Skolem Theorem (DLS). (We will do this eventually in this course.)

Indeed, within the ZFC formalization, we can define the notion of an **elementary sub-**

structure as follows: we say $X \preceq Y$ iff for every "formula"[a] $\varphi(\bar{u})$ and every matching $\bar{a}$ from X, $X \vDash \varphi(\bar{a})$ iff $Y \vDash \varphi(\bar{a})$. You can imagine that with similar handwaving, we can formalize other notions, like DLS.

Once we have done all of this, we can prove that there is some countable X such that $(X, \in) \preceq (V_\alpha, \in)$. Moreover, note that

- if $\alpha \leq \omega$ then $X = V_\alpha$
- if $\alpha \geq \omega + 1$, then $X \subsetneq V_\alpha$ (X is countable but V_α is uncountable).

Observe that $(X, \in)$ is

well-founded: $X \subseteq V_\alpha \to X \in V_{\alpha+1} \subset \mathbf{WF}$

set-like: clear

extensional: we know that V_α is transitive, hence extensional. This is equivalent to the following fact (*within the ZFC-formalism of ZFC*):

$$V_\alpha \vDash \text{"Extensionality Axiom"}$$

Then, since $X \preceq V_\alpha$,

$$X \vDash \text{"Extensionality Axiom"}$$

as well. This is equivalent to $(X, \in)$ being extensional.

We can also argue more directly for extensionality, which asks that any two distinct members of X be told apart by a member of X. Say $b, c \in X$ with $b \neq c$. Since V_α is extensional, some $a \in V_\alpha$ lies in the symmetric difference $b \triangle c$, so that

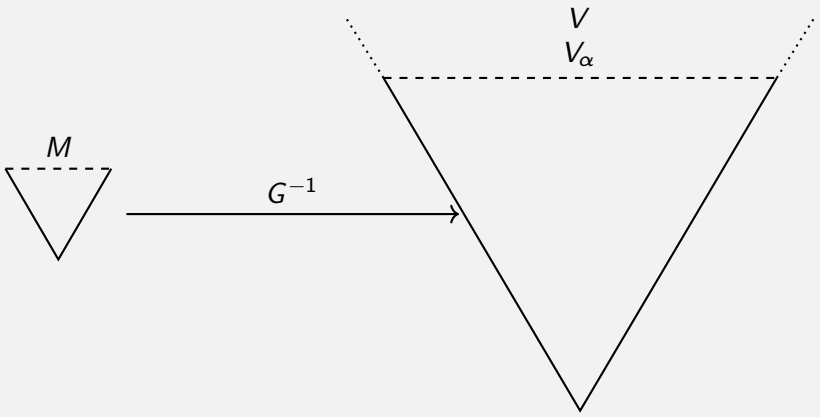
$$V_\alpha \vDash \exists a\,(a \in b \triangle c)$$

This is a "formula" whose parameters b and c both come from X, and $X \preceq V_\alpha$, so X satisfies it too. The witness it supplies is a member of X lying in $b \triangle c$, which is exactly what we wanted.

Let $G : (X, \in) \simeq (M, \in)$ be the Mostowski Collapse Isomorphism (ie, if G is the Mostowski Collapse Isomorphism, then $(M, \in)$ is the Mostowski Collapse—the range of G). As this is an isomorphism, M is transitive. Note that $G^{-1} : M \to V_\alpha$ is an elementary embedding (we compose $G^{-1} : M \simeq X$ with the inclusion $X \hookrightarrow V_\alpha$). In particular, M and V_α satisfy

the same "sentences".

Intuitively, what's going on is that we have built some countable microcosm—some tiny, countable analog—of the entire V-universe, *inside of the ZFC formalism*. This is one of the most important subroutines in set theory: the idea that you can take a big initial piece of V, take some elementary substructure via DLS, take its Mostowski Collapse, and do something with it. We will see a lot of this later on.



[a] There are no formulae because we're in a formalism where everything's a set! But there are "formulae": just make air quotes and change your tone of voice and everything is ok :D

Chapter 3

Relativization

The last two chapters were about sets, classes and the relations on them. This one is about formulae: what it takes for one to be true *in* a class rather than in a set, and how much of the universe a formula has to look at before we can tell.

We built **WF** in Chapter 1 and saw that Foundation says exactly that every set lies in it. To ask whether the rest of the axioms are true in **WF**, we need a notion of truth in a class, and satisfaction will not do: it is a notion about sets. Relativization is what replaces it.

With relativization in hand, we ask which formulae mean the same thing in a transitive class as in V, and which axioms survive the passage to one. The answers give models of large fragments of **ZFC**: the levels V_α, the sets H_κ of hereditarily small sets, and, by reflection, a transitive model of any finitely many axioms of **ZFC**. All of this is what we need to build Gödel's constructible universe L, a transitive class model of all of **ZFC** and **GCH**, which we do in Chapter 4.

3.1 Relativization to a Class

Relativization is essentially like satisfaction, just for classes. Our ultimate goal is the following: given (A, R) a class structure and a formula ψ, we'd want to define $\psi^{(A,R)}$ to be the relativization (think "interpretation") of ψ to (A, R).

Here are some complications with this technique:

- R might not be $\in$
- There might be parameters involved in the definitions of A and R. For instance, if A is a set, then "$A = \{x \mid x \in A\}$" takes in a set A as a parameter.

Look at $A = \{x \mid \varphi(x)\}$. As a scheme, define $\psi^A(\bar{v})$ (with $\bar{v}$ denoting free variables) by recursion on the complexity of $\psi(\bar{v})$ as follows. We will have atomic, propositional and quantifier steps.

Before starting, rename the variables of φ other than x so that none of them occurs in ψ. Write φ' for the result, so that $A = \{x \mid \varphi'(x)\}$ still, and use φ' in place of φ throughout. Both directions of the clash matter. The quantifier step below substitutes a variable u of ψ for the free variable x, so a u bound in φ would be captured and $\varphi(u)$ would say something other than "$u \in A$"; and what that substitution produces is then dropped underneath the quantifiers of ψ, so a parameter of φ sharing its name with a variable that ψ binds would be captured the other way.

- Atomic: We interpret equality and membership symbols as equality and membership. Ie,

$$(v_0 = v_1)^A \text{ is } v_0 = v_1$$
$$(v_0 \in v_1)^A \text{ is } v_0 \in v_1$$

- Propositional: If ψ is a propositional combination of ψ_0 and ψ_1, then ψ^A is the same propositional combination of ψ_0^A and ψ_1^A.
- Quantifiers: We relativize a quantifier by restricting it to A. Ie,

$$(\forall u\, \psi)^A \text{ is } \forall u \left[\varphi'(u) \to \psi^A\right]$$
$$(\exists u\, \psi)^A \text{ is } \exists u \left[\varphi'(u) \wedge \psi^A\right]$$

The atomic step above is written for the case $R = \in$. For a general R it is that step which changes, because "$\in$" is a symbol of the language we are interpreting, and what it is interpreted as in (A, R) is R. So if $R = \{(x, y) \mid \varphi_R(x, y, q)\}$, then $(v_0 \in v_1)^{(A,R)}$ is $\varphi_R(v_0, v_1, q)$, and since the recursion bottoms out at the atomic subformulae, this replaces every occurrence of $\in$ in ψ. The parameter q simply comes along for the ride, which is exactly why φ_R needs the same renaming as φ did, in both directions: in particular q must not be a variable that ψ binds.

3.2 The Relative Consistency of ZF

Why are we doing all of this? Here's one reason.

Theorem 3.2.1 (working in $\mathbf{ZF} - \mathbf{F}$). *For each axiom $\sigma \in \mathbf{ZF}$ (**including** foundation), $\sigma^{\mathbf{WF}}$.*

The above reads, "The relativization of each **ZF** axiom to **WF** is true, working in $\mathbf{ZF} - \mathbf{F}$." Another equivalent formulation is that for all axioms σ of **ZF**,

$$\mathbf{ZF} - \mathbf{F} \vdash \sigma^{\mathbf{WF}} \tag{3.2.1}$$

Before drawing the consequence we are after, we isolate the fact about relativization that the argument will run on. We built ψ^A to be the interpretation of ψ in A, and an interpretation ought to carry proofs and not just statements: reading every quantifier of a proof as ranging over A turns each of its steps into a step about A. The one thing that can go wrong is A being empty, since first-order logic assumes a non-empty domain and proves $\forall u\, \psi \to \exists u\, \psi$; so non-emptiness is carried as a hypothesis rather than as a side condition, being a sentence of the object language like any other.

Lemma 3.2.2 (Scheme). *Let $A = \{x \mid \varphi(x)\}$, say with no parameters, and let σ and τ be sentences. Then, in the meta-theory, whenever $\sigma \vdash \tau$, we have σ^A, $\exists x\, \varphi(x) \vdash \tau^A$.*

Theorem 3.2.1 and Lemma 3.2.2 together yield the following corollary.

Corollary 3.2.3. *If $\mathbf{ZF} - \mathbf{F}$ is consistent, then so is **ZF**.*

Proof. We argue by contraposition. Suppose **ZF** is inconsistent. Say $\sigma_1, \ldots, \sigma_n$ are axioms of $\mathbf{ZF} - \mathbf{F}$ and

$$\{\sigma_1, \ldots, \sigma_n\} \cup \{F\} \vdash \varphi \wedge \neg\varphi$$

Apply Lemma 3.2.2 with σ the conjunction $\sigma_1 \wedge \cdots \wedge \sigma_n \wedge F$ and τ the contradiction. Relativization commutes with the propositional connectives, by the propositional step of the definition, so what

comes out is

$$\left\{\sigma_1^{\mathbf{WF}}, \ldots, \sigma_n^{\mathbf{WF}}\right\} \cup \left\{F^{\mathbf{WF}}\right\} \cup \{\exists x\,(x \in \mathbf{WF})\} \vdash \varphi^{\mathbf{WF}} \wedge \neg\varphi^{\mathbf{WF}}$$

□

Together with the above theorem scheme, we get that **ZF** − **F** is inconsistent—still under the supposition that **ZF** is. Indeed, Theorem 3.2.1 gives $\mathbf{ZF} - \mathbf{F} \vdash \sigma_i^{\mathbf{WF}}$ for each i, and also $\mathbf{ZF} - \mathbf{F} \vdash F^{\mathbf{WF}}$; and $\mathbf{ZF} - \mathbf{F} \vdash \exists x\,(x \in \mathbf{WF})$, since $\emptyset \in V_1$. So every hypothesis on the left of that display is a theorem of **ZF** − **F**, and hence $\mathbf{ZF} - \mathbf{F} \vdash \varphi^{\mathbf{WF}} \wedge \neg\varphi^{\mathbf{WF}}$.

3.3 Complexity of Formulae

Whether a formula says the same thing inside a class as it does in V depends on how its quantifiers range, so we need a way of measuring that. The first step is to give the quantifiers that range over a set, rather than over the whole universe, notation of their own.

Notation (Bounded Quantifiers).

- $\forall u \in v\,\varphi$ stands for $\forall u\,[u \in v \to \varphi]$
- $\exists u \in v\,\varphi$ stands for $\exists u\,[u \in v \wedge \varphi]$

Quantifiers written this way are said to be **bounded**.

A formula all of whose quantifiers are bounded can only ever look inside the sets it is handed, and such formulae sit at the bottom of the hierarchy we are about to build.

Definition 3.3.1 (Δ_0 Formulae). A formula is Δ_0 iff every quantifier occurring in it is bounded.

Everything above the bottom level is obtained by putting a block of unbounded quantifiers, all of the same kind, in front of a Δ_0 formula.

Definition 3.3.2 (Σ_1 and Π_1 Formulae).

- Σ_1 formulae look like

$$\exists u_0 \exists u_1 \cdots \exists u_n \, \varphi$$

  where φ is Δ_0.
- Π_1 formulae look like

$$\forall u_0 \forall u_1 \cdots \forall u_n \, \psi$$

  where ψ is Δ_0.

There are no Δ_1 formulae. On the shapes above, a formula cannot begin with both an unbounded $\exists$ and an unbounded $\forall$; and allowing either block to be empty only puts the Δ_0 formulae into both classes, which buys us nothing we did not already have. What there are is Δ_1^T formulae, and to get them we first have to fix a theory to do the translating for us.

Definition 3.3.3 (Complexity Relative to a Theory). Let T be a theory and let $\psi(\bar{u})$ be a formula. We say $\psi(\bar{u})$ is Δ_0^T iff there is a Δ_0 formula $\varphi(\bar{u})$ such that

$$T \vdash \forall \bar{u} \, [\psi(\bar{u}) \leftrightarrow \varphi(\bar{u})]$$

Similarly for Σ_1^T and Π_1^T. We say $\psi(\bar{u})$ is Δ_1^T iff it is both Σ_1^T and Π_1^T.

Keep going to get Σ_n and Π_n formulae, and also Σ_n^T, Π_n^T and Δ_n^T formulae.

3.4 Absoluteness

3.4.1 Absoluteness Theorems

We will constantly be comparing what a formula says in a smaller class with what it says in a bigger one, so we give the possible outcomes names.

Definition 3.4.1 (Absoluteness). Let $M \subseteq N$ be classes and let $\varphi(\bar{u})$ be a formula. We say $\varphi(\bar{u})$ is

- **downwards absolute** from N to M iff for all $\bar{x} \in M^{<\omega}$ matching $\bar{u}$, $\varphi^N(\bar{x}) \to \varphi^M(\bar{x})$.

- **upwards absolute** from M to N iff for all $\bar{x} \in M^{<\omega}$ matching $\bar{u}$, $\varphi^M(\bar{x}) \to \varphi^N(\bar{x})$.
- **absolute** between M and N iff it is both, ie, iff for all $\bar{x} \in M^{<\omega}$ matching $\bar{u}$, $\varphi^M(\bar{x}) \leftrightarrow \varphi^N(\bar{x})$.

When $N = V$, we just say $\varphi(\bar{u})$ is downwards absolute, upwards absolute or absolute **for** M.

We will want to relativize classes as well as formulae, and it is convenient to have notation for it.

Notation. Say M is a transitive class and A another class. Say $A = \{x \mid \varphi(x)\}$. We might write A^M to mean $\left\{x \in M \;\middle|\; \varphi^M(x)\right\}$ even though it may depend on the choice of φ with this notation.

A^M is what M thinks A is. When it agrees with $A \cap M$, ie, when $\varphi(x)$ is absolute for M, we say A is **absolute** for M.

We will use this constantly: in Section 3.4.3 we ask whether a class function G defined by recursion satisfies $G^M = G \restriction (A \cap M)$, and in Section 3.4.4 whether $\mathbf{OR}^M = \mathbf{OR} \cap M$.

Every absoluteness theorem we prove will have an assumption that the smaller model (or sometimes, both models) are transitive. There are no absoluteness theorems in this course that do not mention transitivity, and it is a common mistake to try and apply them in cases where there isn't transitivity.

Lemma 3.4.2 (Δ_0-Absoluteness). *Let M and N be classes[a], with $M \subseteq N$. Assume M is transitive. Then, for every Δ_0 formula $\varphi(\bar{u})$ and $\bar{x} \in M^{<\omega}$ (matching $\bar{u}$ in length),*

$$\varphi^M(\bar{x}) \leftrightarrow \varphi^N(\bar{x})$$

That is, $\varphi(\bar{u})$ is absolute between M and N.

[a] possibly proper classes, possibly sets

Proof. We argue by induction on the complexity of $\varphi(\bar{u})$.

- Atomic Case. Note that

$$(u_0 = u_1)^M \qquad u_0 = u_1 \qquad (u_0 = u_1)^N$$

are all exactly the same. Similarly,

$$(u_0 \in u_1)^M \qquad u_0 \in u_1 \qquad (u_0 \in u_1)^N$$

are all exactly the same.

- Propositional Case. If φ is some propositional combination of φ_0 and φ_1, then the statement for φ is clear from the induction hypotheses for φ_0 and φ_1.

- Bounded Quantifiers. Say $\varphi(\bar{u})$ is of the form $\exists v \in w\ (\psi(\bar{u}, v, w))$. (This is enough, because bounded universal can be formulated as a negation of a bounded existential, and the propositional case covers negations.)

  Let $\bar{x} \in M^{<\omega}$ and $z \in M$. Then, the following are equivalent:

  - $[\exists v \in z,\ \psi(\bar{x}, v, z)]^M$
  - There is $y \in z \cap M$ such that $\psi^M(\bar{x}, y, z)$
  - There is $y \in z$ such that $\psi^M(\bar{x}, y, z)$ (because M is transitive, so $z = z \cap M$)
  - There is $y \in z \cap N$ such that $\psi^M(\bar{x}, y, z)$ (because $z \subseteq M \subseteq N$, so $z = z \cap N$)
  - There is $y \in z \cap N$ such that $\psi^N(\bar{x}, y, z)$
  - $[\exists v \in z,\ \psi(\bar{x}, v, z)]^N$

□

One level up, we only get one direction each.

Lemma 3.4.3. *In the same situation as above, we have upward absoluteness from M to N for Σ_1 formulae and downward absoluteness from N to M for Π_1 formulae.*

Relative to a theory, the same holds for formulae that the theory proves equivalent to these shapes.

Lemma 3.4.4. *Let T be a theory and $M \subseteq N$ be models of T (with M transitive). Then,*

- Σ_1^T *formulae are upwards-absolute*
- Π_1^T *formulae are downwards-absolute*
- Δ_1^T *formulae are absolute*

3.4.2 Finite Sets

For the rest of this section, let M be a transitive class model of $\mathbf{ZF} - \mathbf{P}$. We start with the finite sets.

Lemma 3.4.5. $[M]^{<\omega} \subseteq M$.

Proof. Given $x_1, \ldots, x_n \in M$, $\{x_1, \ldots, x_n\} = \bigcup_{j=1}^{n} \{x_j\}$. □

The formula "A is finite" is absolute for M. In fact, we can do better.

Lemma 3.4.6. *For each $n < \omega$, the formula "$|A| = n$" is absolute for M.*

Proof.

Upwards absoluteness. Say $f \in M$ and

$$\left(f : n \xrightarrow{\sim} A\right)^M$$

where $f : n \xrightarrow{\sim} A$ means f is injective and surjective.

We know that $f : n \xrightarrow{\sim} A$ because we already know it is absolute.

Downwards absoluteness. We have that $f : n \xrightarrow{\sim} A$ in V. But $f \subseteq_{\text{finite}} n \times A \subseteq M$. And indeed we've checked already that Cartesian products are absolute.

Then, by the previous lemma, $f \in M$.

Now again, by Δ_0-absoluteness,

$$\left(f : n \xrightarrow{\sim} A\right)^M$$

□

Being finite and having a given finite size are the same thing, so we get the following for free.

Lemma 3.4.7. *"A infinite" is absolute for M.*

Proof. Clear from previous. □

The power set of a finite set is also computed correctly.

Lemma 3.4.8. *If $A \in M$ and A is finite (in both M and V), then $\mathcal{P}(A) \in M$. Therefore, $A \mapsto \mathcal{P}(A)$, restricted to finite sets A, is an absolute class function for M.*

For the remainder of this argument, let's work in M.

We argue by recursion. We have that A is finite, so there is some $n < \omega$ and $f : n \xrightarrow{\sim} A$. Denote by 2^n the set of all n-tuples of 0s and 1s. For each $x \in 2^n$, write

$$B_x = \{f(i) \mid x_i = 1\} \subseteq A$$

In particular, we have a mapping $x \mapsto B_x$. Finally, we can conclude that

$$\mathcal{P}(A) = \{B_x \mid x \in 2^n\}$$

3.4.3 Absoluteness of Recursive Definitions

Often, we have (A, R) well-founded and set-like, and some $F : A \times V \to V$. We then define $G : A \to V$ by recursion so that

$$\forall y \in A,\, G(y) = F(y, G \restriction \{x \mid x\, R\, y\})$$

Now, we'd like to know that if the "data" above is absolute, then so is G. How do we show this?

Ok, let's list our assumptions (recalling that a class function is absolute iff the defining formula is absolute, and also M is closed under the values):

- A and R are absolute
- F is an absolute class function
- $y \mapsto \{x \mid x\, R\, y\}$ is an absolute class function

We can conclude from this the following two lemmas.

Lemma 3.4.9. $[(A, R)$ *is set-like and well-founded*$]^M$

Proof sketch. Note that (A, R) is well-founded iff

$$\forall S,\, [\emptyset \neq S \subseteq A \to \exists x \in S,\, \forall y \in S,\, \neg (y\, R\, x)]$$

We can essentially rewrite the expression inside brackets using quantifiers that are bounded over (A, R) (subject to the universal $\forall S$). The bracketed expression is absolute between M and V, and the universal out front makes it downwards absolute from V to M.

Note that set-like is downwards absolute from V to M because we assumed that $y \mapsto \{x \mid x\, R\, y\}$ and (A, R) are all absolute. □

With that, G is computed correctly in M.

Lemma 3.4.10. *G is absolute for M. In particular, for every $y \in A^M$,*

$$G^M(y) = F^M\big(y, G^M \restriction \{x \mid x\, R^M\, y\}\big)$$

and

$$G^M = G \restriction A^M = G \restriction (A \cap M)$$

Proof sketch. Just show that for all y in $A^M = A \cap M$, we have $G^M(y) = G(y)$ by induction on y. □

There are many useful applications of absoluteness.

Example 3.4.11 (Defining Ordinal Arithmetic)**.** There are many ways we can use absoluteness to define ordinal arithmetic.

- Defining it recursively. For example, we can define $\alpha+0=\alpha$, $\alpha+(\beta+1)=(\alpha+\beta)+1$, and if γ is a limit ordinal, we can define $\alpha+\gamma:=\sup_{\beta<\gamma}(\alpha+\beta)$.
- Defining it using Mostowski Collapse of well-orderings. For example, we can define

$$\alpha+\beta:=\text{type}((\{0\}\times\alpha)\cup(\{1\}\times\beta),<_{\text{lex}})$$

  Note that type is the range of the Mostowski Collapse. Indeed,

$$G(y)=\{G(x)\mid x<_{\text{lex}}y\}$$

  is absolute.

3.4.4 Ordinals in Transitive Models

If M is a transitive model of $\mathbf{ZF}-\mathbf{P}$, we claim that the statement "α is an ordinal" is absolute for M.

But what does it even mean? It means

$$\mathbf{OR}^M=\mathbf{OR}\cap M$$

As M is transitive, $\mathbf{OR}^M$ is transitive, so either $\mathbf{OR}^M\in\mathbf{OR}$ or $\mathbf{OR}^M=\mathbf{OR}$. It turns out:

1. $\mathbf{OR}^M\in\mathbf{OR}$ iff M is a set
2. $\mathbf{OR}^M=\mathbf{OR}$ iff M is a proper class

To see this, note that foundation is equivalent to $(V,\in)$ being well-founded. It's clear that $(V,\in)$ is set-like: if y is a set, then everything that is $\in$-below y is just y, which is a set. So by the Theorem Scheme on Recursion (which is true in $\mathbf{ZF}-\mathbf{P}$), we have a class function $G:V\to\mathbf{OR}$ such that $\forall y,\ G(y)=\sup_{x\in y}(G(x)+1)$.

Now if we also give ourselves the power set axiom, then we can define the V-hierarchy, and conclude

that $G(y) = \text{rank}(y)$. But without power sets, *we cannot even define the V-hierarchy*, so there is no meaningful notion of rank!

By the theorem scheme on absoluteness of recursively defined class functions, G is an absolute class function for M. In other words, $G^M = G \restriction M$. Also, $G^M : M \to \mathbf{OR}^M$ is surjective. Indeed, each value of G^M is an ordinal in M, and $G(\alpha) = \alpha$ for every ordinal α, by induction: $G(\alpha) = \sup_{\beta<\alpha}(G(\beta)+1) = \sup_{\beta<\alpha}(\beta+1) = \alpha$. So every $\alpha \in \mathbf{OR}^M = \mathbf{OR} \cap M$ is $G^M(\alpha)$.

This gives both halves of the dichotomy.

- If M is a set, then $\mathbf{OR} \cap M = G[M]$ is a set by Replacement, and a transitive set of ordinals is an ordinal, so $\mathbf{OR}^M \in \mathbf{OR}$.
- If M is a proper class, suppose $\mathbf{OR} \cap M$ were a set, hence an ordinal α. In V we do have Power Set, so $G(y) = \text{rank}(y)$ for every y. So, for every $y \in M$, $\text{rank}(y) = G^M(y) \in \mathbf{OR} \cap M = \alpha$, ie, $y \in V_\alpha$. Then $M \subseteq V_\alpha$, and M would be a set by Comprehension. So $\mathbf{OR}^M$ is a transitive proper class of ordinals, which can only be $\mathbf{OR}$.

Now let's study the uncountable ordinals in M.

Denote by ω_1^M what "M thinks" is the least uncountable ordinal. If $(\omega_1 \text{ exists})^M$, ie, if the notion of the existence of the least uncountable ordinal is even definable in M, then $\omega_1^M \leq \omega_1$ (where $\leq$ is just the usual $\leq$ on ordinals—recall that $\mathbf{OR}$ is absolute, which means precisely that whatever a class *thinks* is an ordinal is, indeed, an ordinal in V).

3.5 Relativizing the Axioms of ZFC

We now put absoluteness to work on the axioms themselves, asking which of them hold relativized to which transitive classes. This is what Theorem 3.2.1 needed for **WF**, and it also finds models of large fragments of **ZFC** among the levels of the V-hierarchy.

3.5.1 The First Few Axioms

For this discussion, fix a class M. Let's list out the axioms of $\mathbf{ZF} - \mathbf{F}$, relativized to M.

We start with Extensionality, which relativized to M reads

$$[\forall A \forall B\, [\forall x\, (x \in A \leftrightarrow x \in B) \to A = B]]^M$$

Or, equivalently,

$$[\forall A \forall B\, [(A \subseteq B \wedge B \subseteq A) \to A = B]]^M$$

where $A \subseteq B$ is the Δ_0 formula $(\forall x \in A)\,(x \in B)$. Indeed, all sub-formulae of the above are Δ_0.

This is all we need, as long as M is transitive.

Lemma 3.5.1 (Extensionality). *If M is transitive, then (Extensionality)^M.*

Proof. By Δ_0-absoluteness, for $A, B \in M$ the formula $(A \subseteq B \wedge B \subseteq A) \to A = B$ says the same thing in M as it does in V, where it is true. So it holds in M for all $A, B \in M$, which is exactly (Extensionality)^M. □

From now on, assume that M is transitive, so that (Extensionality)^M.

Comprehension needs M to contain enough subsets of its members.

Lemma 3.5.2 (Comprehension). *If M is closed under subsets, ie, $A \in M$ and $B \subseteq A$ imply $B \in M$, then (Comprehension)^M.*

Proof. For each formula $\varphi(x, \bar{y})$, the corresponding instance relativized to M is

$$\forall A \in M\, \forall \bar{y} \in M\, \exists B \in M\, \forall x \in M\, \left[x \in B \leftrightarrow (x \in A \wedge \varphi^M(x, \bar{y}))\right]$$

As M is transitive, any such B and the given A are subsets of M, so the bracket pins B down completely: it must be $\{x \in A \mid \varphi^M(x, \bar{y})\}$. So Comprehension relativizes to M iff each of these sets lies in M, and in particular it does whenever M is closed under subsets. □

From now on, assume (Comprehension)^M as well.

Pairing only needs M to be closed under pairs.

Lemma 3.5.3 (Pairing). *If $\{x, y\} \in M$ for all $x, y \in M$, then $(\mathit{Pairing})^M$.*

Proof. Relativized to M, Pairing says

$$\forall x \in M \, \forall y \in M \, \exists A \in M \; (x \in A \wedge y \in A)$$

If $\{x, y\} \in M$, then it is a witness. (Here we did not even need transitivity, because the formula under the quantifiers is atomic.) □

3.5.2 The Levels of the V-Hierarchy

Let's see which axioms hold in the levels of the V-hierarchy.

Proposition 3.5.4. *For each axiom σ of {Extensionality, Comprehension, Pairing, Unions, Power Sets}, for all limit ordinals α, σ^{V_α}.*

At V_ω, Replacement holds as well.

Proposition 3.5.5. *For each axiom σ of {Extensionality, Comprehension, Pairing, Unions, Power Sets, Repla σ^{V_ω}.*

Above V_ω, a limit level need not satisfy Replacement.

Non-Example 3.5.6 ($V_{\omega+\omega}$). We can show that Replacement does not relativize to $V_{\omega+\omega}$. Indeed, consider the class function $G : n \mapsto \omega + n$. It's definable in two ways:

1. $G(n) = S$ iff $\exists g \, [g$ is a function $\wedge \, \mathrm{dom}(g) = n + 1 \wedge g(0) = \omega \wedge \forall i < n \; (g(i+1) = g(i) \cup \{g(i)\})$
2. $G(n) = S$ iff $\forall g \, [(g$ is a function $\wedge \, \mathrm{dom}(g) = n + 1 \wedge g(0) = \omega \wedge \forall i < n \; (g(i+1) = g(i) \cup \{g(i)\})$

Both say that S is the last value of the function g on $n+1$ that starts at ω and takes a successor at each step, and in both, everything after the unbounded quantifier over g is Δ_0 in g, n, S and the parameter ω. So the first is Σ_1 and the second is Π_1.

Now fix $n < \omega$. The unique g in question is $\{(i, \omega + i) \mid i \leq n\}$, and each of its pairs lies in $V_{\omega+n+3}$, so $g \in V_{\omega+n+4} \subseteq V_{\omega+\omega}$. Hence, for $S \in V_{\omega+\omega}$, the first formula relativized to $V_{\omega+\omega}$ holds iff $S = \omega + n$: the witness g is available inside $V_{\omega+\omega}$, and the Δ_0 part means the same

thing there as in V by Δ_0-absoluteness. (The second formula relativizes correctly too, by downward absoluteness of Π_1 formulae, so it does not matter which definition we use.)
So the hypothesis of the instance of Replacement for this formula, with $A = \omega \in V_{\omega+\omega}$ and parameter ω, holds in $V_{\omega+\omega}$. If the conclusion did too, we would have $B \in V_{\omega+\omega}$ with $\omega + n \in B$ for every $n < \omega$. But $B \in V_{\omega+k+1}$ for some $k < \omega$, so $B \subseteq V_{\omega+k}$, and then $\omega + k \in V_{\omega+k}$, ie, $\text{rank}(\omega + k) < \omega + k$, which is absurd since ordinals are their own ranks.

We now continue going through the axioms.

Let's look at Foundation.

We want to see if $\mathbf{F}^M$, that is,

$$[\forall A\,(A \neq \emptyset \to \exists x \in A,\, \forall y \in A,\, y \notin x)]^M$$

Consider the statement

$$\forall A \in M\,(A \neq \emptyset \to \exists x \in A,\, \forall y \in A,\, y \notin x)$$

This is true if $M \subseteq \mathbf{WF}$.

We know the following:

- if $A \in \mathbf{WF}$, then $A \subset \mathbf{WF}$
- if $x \in A$ with $\text{rank}(x) = \min\{\text{rank}(y) \mid y \in A\}$ then x is $\in$-minimal in A.
- if $y \in x$ then $\text{rank}(y) < \text{rank}(x)$

In particular, $\mathbf{F}^{\mathbf{WF}}$ and $\mathbf{F}^{V_\alpha}$.

Finally, for infinity, we can show that if $\omega \in M$, then infinity relativizes to M; in particular, infinity relativizes to $\mathbf{WF}$ and to V_α for $\alpha > \omega$ a limit.

We're now done showing that every axiom of $\mathbf{ZF}$ without infinity relativizes to V_ω, and furthermore, that every axiom of $\mathbf{ZF}$ relativizes to $\mathbf{WF}$. As we saw earlier, this is what gives us Corollary 3.2.3: if $\mathbf{ZF} - \mathbf{F}$ is consistent, then so is $\mathbf{ZF}$.

3.5.3 Choice

Now, let's study choice. We know that under $\mathbf{ZF} - \mathbf{F}$, this is equivalent to the well-ordering principle.

We want to show, effectively, that

$$[\forall A \exists R\,((A, R) \text{ is a w.o.})]^M$$

This follows from

$$\forall A \in M,\, \exists R \in M\ ((A, R) \text{ is a w.o.})$$

because the statement "(A, R) is a w.o." is Π_1, and thus, downwards absolute.

Indeed, the reason this statement is Π_1 is that (A, R) is a w.o. iff (A, R) is a l.o. (which is a Δ_0 statement) and (A, R) is well-founded (which is Π_1 because it says $\forall S\,[(\emptyset \neq S \subseteq A) \to \exists x \in S\, \forall y \in S \neg (y\, R\, x)$ and everything after $\forall S$ is Δ_0: the quantifiers over x and y are bounded by S, and $\neg (y\, R\, x)$ says that the ordered pair (y, x) is not a member of R, which needs only quantifiers bounded by R).

In the background, now, assume $\mathbf{ZFC} - \mathbf{F}$. Let $A \in \mathbf{WF}$ and suppose $A \subseteq V_\alpha$ for some α. In V, we have some $R \subseteq A \times A$ such that R is a w.o. Then $R \subseteq V_\alpha \times V_\alpha$, so $R \in V_{\alpha+3} \subset \mathbf{WF}$. By Π_1 downwards absoluteness, we conclude that $[(A, R) \text{ is a w.o.}]^{\mathbf{WF}}$.

Corollary 3.5.7. $(\mathbf{AC})^{\mathbf{WF}}$, $(\mathbf{AC})^{\mathbf{V}_\alpha}$ *for all limit* α.

Since **WF** satisfies Choice, the relative consistency result of Corollary 3.2.3 extends to **ZFC**.

Corollary 3.5.8. *If* $\mathbf{ZFC} - \mathbf{F}$ *is consistent, then so is* **ZFC**.

Being a well-ordering is in fact also Σ_1, over a weak enough theory, so it is absolute for transitive models of that theory and not merely downwards absolute.

Lemma 3.5.9. *If we denote by* T *the theory consisting of* **ZF** *without power sets and infinity, the statement "*(A, R) *is a w.o." is also* Σ_1^T.

Proof. (A, R) is a w.o. iff there is some $F : A \to \mathbf{OR}$ such that $\forall x, y \in A\,(x\,R\,y \to F(x) < F(y))$ and (A, R) is a l.o. From left to right, take F to be the collapsing map of Theorem 1.2.4, which is order-preserving with ordinal values; its construction is a recursion along a well-ordering of a set, and uses Replacement but neither Power Set nor Infinity. From right to left, given a nonempty $S \subseteq A$, the set $F[S]$ is a nonempty set of ordinals, so it has a least element $F(x)$ with $x \in S$, and no $y \in S$ has $y\,R\,x$, as that would give $F(y) < F(x)$.

So, in T, "(A, R) is a w.o." is equivalent to

(A, R) is a l.o. $\wedge\, \exists F\,[F$ is a function $\wedge\, \mathrm{dom}(F) = A \wedge \forall x \in A\ (F(x) \in \mathbf{OR}) \wedge \forall x, y \in A\ (x\,R\,y \to F(x) \in F($

Everything after $\exists F$ is Δ_0 provided "$F(x) \in \mathbf{OR}$" is, and this is where Foundation is used: in its presence, an ordinal is just a transitive set that is linearly ordered by $\in$, which is a Δ_0 condition. □

3.5.4 Inaccessible Cardinals

To get a level of the V-hierarchy satisfying all of **ZFC**, we need a level that is closed under everything, and that calls for some definitions about cardinals.

Definition 3.5.10 ((Strong) Limit Cardinals). μ is a

- **limit cardinal** iff for all cardinals $\kappa < \mu$, $\kappa^+ < \mu$.
- **strong limit cardinal** iff for all cardinals $\kappa < \mu$, $2^\kappa < \mu$.

A cardinal can be a limit while still being reachable in fewer steps than its size, which is what cofinality measures.

Definition 3.5.11 (Cofinality). The **cofinality** of an ordinal A, denoted $\mathrm{cf}(A)$, is the least $|B|$ such that there is a function $f : B \to A$ whose image $f[B]$ is unbounded in A.

The cardinals that cannot be reached in fewer steps get a name.

Definition 3.5.12 (Regularity/Singularity). A cardinal μ is **regular** if $\mathrm{cf}(\mu) = \mu$. If μ is not regular, we say μ is **singular**.

Putting these together gives the cardinals we are after.

Definition 3.5.13 (Weak/Strong Inaccessibility). We say a cardinal μ is

- **weakly inaccessible** iff μ is an uncountable, regular limit cardinal.
- **strongly inaccessible** iff μ is an uncountable, regular, strong limit cardinal.

A strongly inaccessible cardinal is a fixed point of the $\beth$ function.

Proposition 3.5.14. *If μ is strongly inaccessible, then $\beth_\mu = \mu$.*

Proof. By induction, $\beth_\alpha \geq \alpha$ for every ordinal α, so $\beth_\mu \geq \mu$.

It's enough to show that $\forall \alpha < \mu$, $\beth_\alpha < \mu$, because the fact that μ is a limit ordinal means that, by definition, $\beth_\mu = \sup_{\alpha<\mu} \beth_\alpha$.

Now perform induction on $\alpha < \mu$.

Base case: $\beth_0 < \mu$ because $\beth_0 = \omega$ and μ is uncountable.

Successor case: if $\alpha + 1 < \mu$ and $\beth_\alpha < \mu$, then $\beth_{\alpha+1} = 2^{\beth_\alpha} < \mu$, because $\beth_\alpha$ is a cardinal below μ and μ is a strong limit.

Limit case: if $\alpha < \mu$ is a limit and $\beth_\gamma < \mu$ for all $\gamma < \alpha$, then $\beth_\alpha = \sup_{\gamma<\alpha} \beth_\gamma$ is a supremum of $|\alpha| < \mu$ many ordinals below μ, so it is below μ, as μ is regular. □

With these, all the axioms hold at once.

Corollary 3.5.15. *If M is strongly inaccessible, then σ^{V_M} for all axioms σ of* **ZFC**.

3.6 Relativization vs Satisfaction

Can we formalize into Set Theory the notion of "$M \vDash \varphi(\bar{x})$"?

Pick "symbols" and formalize the following using **ZF** − **P**:

- syntax of first-order logic

- definition of satisfaction
- proof theory
- completeness theorem

The way we define what "formulae" are internally is using Gödel numbers. Ie, for each formula φ (*external*), there exists some set $\ulcorner\varphi\urcorner$ that is analogous to it. We still write $M \vDash \varphi(\bar{x})$.

It is possible to show that our analogies translate both ways, ie, $\varphi(\bar{x})^M$ iff "$M \vDash \varphi(\bar{x})$", simply by structural induction on φ.

Note, about $M \vDash \varphi(\bar{x})$:

1. φ is not a set. There's a set $\ulcorner\varphi\urcorner$ that is analogous to φ. Nevertheless, we just write $M \vDash \varphi(\bar{x})$.
2. This requires $M \neq \emptyset$ and $\bar{x} \in M^{n+1}$, where $v_0, \ldots, v_n$ are the free variables of φ. Contrast with $\varphi^M(\bar{x})$.
3. $\varphi^M(\bar{x})$ was defined for each φ and each class M. $M \vDash \varphi(\bar{x})$ is defined only for sets M:

$$\vDash \subseteq \{(M, c, \bar{x}) \mid M \neq \emptyset,\ c \text{ a Gödel number}, \bar{x} \in M^{<\omega}\}$$

These agree wherever both make sense.

Lemma 3.6.1 (Fact (Scheme)). *Consider any formula $\varphi(v_0, \ldots, v_n)$. For each $M \neq \emptyset$ and $\bar{x} \in M^{n+1}$,*

$$\varphi(\bar{x})^M \text{ iff } \text{“}M \vDash \varphi(\bar{x})\text{”}$$

For now, work in **ZF**. Given a formula $\varphi(u)$ and a set x, say $\varphi(u)$ **defines** x iff

$$\forall u\,[u = x \leftrightarrow \varphi(u)] \tag{$\star$}$$

Observe: there is no formula $\delta(r, x)$ so that, for every formula $\varphi(u)$,

$$\forall x \left[\delta(\ulcorner \varphi(u) \urcorner, x) \leftrightarrow \underbrace{\varphi(u) \text{ defines } x}_{(\star)} \right]$$

Otherwise, the least undefinable ordinal would exist and be definable (because there are only countably many Gödel numbers).

3.7 The Sets H_κ

The levels V_α are cut out by rank. We now cut out sets by size instead—hereditarily, so that a set is small only if everything below it is. We will work in **ZFC** because we need choice to be able to work comfortably with cardinalities.

3.7.1 Hereditary Cardinality

We measure a set by the size of its transitive closure.

Definition 3.7.1 (H_κ)**.** For each cardinal $\kappa \geq \omega$, define the class

$$H_\kappa = \{z \mid |\mathrm{trcl}(\{z\})| < \kappa\}$$

Note that

$$\mathrm{trcl}(\{z\}) = \{z\} \cup \mathrm{trcl}(z)$$

so

$$|\mathrm{trcl}(\{z\})| = 1 \oplus |\mathrm{trcl}(z)|$$

We can also unwind $\mathrm{trcl}(z)$ one level further, through the members of z.

Proposition 3.7.2. *For all sets z,*

$$\mathrm{trcl}(\{z\}) = \{z\} \cup \bigcup_{x \in z} \mathrm{trcl}(\{x\})$$

Nothing is assumed about z here: it need not be transitive, let alone an ordinal.

Lemma 3.7.3. *Each H_κ is a transitive class and, if $\kappa < \lambda$, then $H_\kappa \subsetneq H_\lambda$.*

Proof.

Transitivity. Let $z \in H_\kappa$ and $y \in z$. Then $y \in \mathrm{trcl}(\{z\})$, so $\mathrm{trcl}(\{z\})$ is a transitive set with y as a member, and so it contains the smallest such set: $\mathrm{trcl}(\{y\}) \subseteq \mathrm{trcl}(\{z\})$. Hence $|\mathrm{trcl}(\{y\})| \leq |\mathrm{trcl}(\{z\})| < \kappa$, and $y \in H_\kappa$.

Monotonicity. If $\kappa < \lambda$, then $|\mathrm{trcl}(\{z\})| < \kappa$ implies $|\mathrm{trcl}(\{z\})| < \lambda$, so $H_\kappa \subseteq H_\lambda$. The inclusion is strict because $\kappa \in H_\lambda \setminus H_\kappa$: $\mathrm{trcl}(\{\kappa\}) = \kappa + 1$ (it is transitive and has κ as a member, and any transitive set with κ as a member contains κ and all its members), and $|\kappa + 1| = \kappa$, as κ is infinite, which is below λ but not below κ.

□

In particular, the members of H_κ have small rank.

Lemma 3.7.4. *For all κ, $H_\kappa \subseteq V_\kappa$.*

Proof. Let $z \in H_\kappa$. We will show that $z \in V_\kappa$. Let

$$\sigma = \{\mathrm{rank}(y) \mid y \in \mathrm{trcl}(\{z\})\}$$

Then σ is a set of ordinals and $\mathrm{rank}(z) = \max(\sigma)$.

We claim that σ is a transitive set. Indeed, if not, then we would have some $\alpha < \beta$ such that $\beta = \min(\sigma \setminus \alpha)$. Pick $y \in \mathrm{trcl}(\{z\})$ such that $\mathrm{rank}(y) = \beta$. Then, $\forall x \in y$, since $x \in \mathrm{trcl}(\{y\})$ and $\mathrm{rank}(x) < \mathrm{rank}(y)$, we conclude that $\mathrm{rank}(x) < \alpha$. Then, by the rank formula,

$$\beta = \mathrm{rank}(y) = \sup_{x \in y}(\mathrm{rank}(x) + 1) \leq \alpha$$

This is a contradiction! So σ is transitive as desired.

This then tells us that $\sigma = \mathrm{rank}(z) + 1$: a transitive set of ordinals is an ordinal, and an ordinal

whose largest element is $\text{rank}(z)$ is $\text{rank}(z)+1$. Then, we get

$$|\sigma| \leq \sigma < |\sigma|^+ \leq |\text{trcl}(\{z\})|^+ \leq \kappa$$

so $\text{rank}(z) < \kappa$. □

So H_κ is cut out of a set.

Corollary 3.7.5. *Each H_κ is a set.*

Proof. Just apply comprehension. □

The first two of these get names of their own.

Definition 3.7.6. Define $\mathbf{HF} := H_\omega$ and $\mathbf{HC} := H_{\omega_1}$

We will show that $\mathbf{HC}$ is a model of $\mathbf{ZF} - \mathbf{P}$. In fact, so is H_{κ^+} for each cardinal κ.

Lemma 3.7.7. $\mathbf{HF} = V_\omega$

Proof. We just saw that $\mathbf{HF} \subseteq V_\omega$. Given some $z \in V_\omega$, pick $n < \omega$ such that $z \in V_n$. Then

$$|\text{trcl}(\{z\})| \leq |V_n| < \omega$$

so $z \in \mathbf{HF}$. □

The ordinals in H_κ are the ones we would expect.

Lemma 3.7.8. $\mathbf{OR} \cap H_\kappa = \kappa$

Proof. If $\alpha \in \mathbf{OR}$, then $\text{trcl}(\{\alpha\}) = \alpha + 1$, because $\alpha + 1 = \alpha \cup \{\alpha\}$ is transitive and has α as a member, and any transitive set with α as a member contains α and all its members. So $\alpha \in H_\kappa$ iff $|\alpha + 1| < \kappa$.

If $\alpha < \kappa$, then $\alpha + 1 < \kappa$ too, because κ is an infinite cardinal and so a limit ordinal; so

$|\alpha+1| \leq \alpha+1 < \kappa$. If $\alpha \geq \kappa$, then $|\alpha+1| \geq |\kappa| = \kappa$. So the ordinals in H_κ are exactly those below κ. □

So H_κ and V_α can only coincide when the indices do.

Lemma 3.7.9. *If $H_\kappa = V_\alpha$, then $\kappa = \alpha$.*

Here is a useful lemma.

Lemma 3.7.10. *If κ is regular, then*

$$H_\kappa = \{z \mid \forall x \in \mathrm{trcl}(\{z\}),\ |x| < \kappa\}$$

Proof. Call the RHS I. We show $H_\kappa = I$.

$\underline{H_\kappa \subseteq I}$. Let $z \in H_\kappa$ and $x \in \mathrm{trcl}(\{z\})$. Then $x \subseteq \mathrm{trcl}(\{z\})$ and $|x| \leq |\mathrm{trcl}(\{z\})| < \kappa$.

$\underline{H_\kappa \supseteq I}$. We apply $\in$-induction. Ie, we assume that for $x \in z$, since $x \in I$ (by transitivity of I), $x \in H_\kappa$, and therefore, $|\mathrm{trcl}(\{x\})| < \kappa$.

Performing the induction, we calculate

$$|\mathrm{trcl}(\{z\})| \leq 1 \oplus \bigoplus_{x \in z} \underbrace{|\mathrm{trcl}(\{x\})|}_{<\kappa}$$

As $z \in I$, $|z| < \kappa$. So we are summing $< \kappa$ many cardinals that are all $< \kappa$, resulting in something $< \kappa$.

□

3.7.2 The Axioms in H_κ

So now let's ask ourselves what Axioms of ZFC hold in these H_κ.

Lemma 3.7.11. *The following ZFC axioms hold in all H_κ:*

1. *Extensionality*

2. *Comprehension*
3. *Pairing*
4. *Union*
5. *Choice*
6. *Foundation (if we assume foundation in V)*

Proof.

1. H_κ is transitive.
2. If $A \in H_\kappa$ and $B \subseteq A$, then $B \in H_\kappa$.
3. If $x, y \in H_\kappa$, then $\{x, y\} \in H_\kappa$, since $\mathrm{trcl}(\{\{x, y\}\}) = \{\{x, y\}\} \cup \mathrm{trcl}(\{x\}) \cup \mathrm{trcl}(\{y\})$ has cardinality below κ, as κ is infinite.
4. If $A \in H_\kappa$, then $\bigcup A \in H_\kappa$, since $\bigcup A \subseteq \mathrm{trcl}(A)$, and so $\mathrm{trcl}(\{\bigcup A\}) \subseteq \{\bigcup A\} \cup \mathrm{trcl}(A)$.
5. Let $A \in H_\kappa$. Pick a well-ordering R of A (assuming AC in V). Then, as $R \subseteq A \times A$, $R \in H_\kappa$. Use downwards Π_1-absoluteness to see that

$$[(A, R) \text{ is a well-ordering}]^{H_\kappa}$$

6. `sorry`

□

Infinity needs κ to be large enough to contain ω.

Lemma 3.7.12. *Infinity holds in H_κ iff $\kappa > \omega$.*

Proof. Follows from the fact that $\kappa > \omega \leftrightarrow \omega \in H_\kappa$. □

Replacement is where the regularity of κ comes in.

Lemma 3.7.13. *If $\kappa = \mathrm{cf}(\kappa)$ (ie, κ is regular), then each axiom σ of the Replacement scheme holds relativized to H_κ, ie, σ^{H_κ}.*

Proof. If $A \in H_\kappa$ and $f : A \to H_\kappa$, then, letting $B = \mathrm{range}(f)$, we have

$$\begin{aligned} |\mathrm{trcl}(\{B\})| &= \left| \{B\} \cup \bigcup_{x \in A} \mathrm{trcl}(\{f(x)\}) \right| \\ &\leq 1 \oplus \underbrace{\sum_{x \in A}}_{\text{fewer than } \kappa \text{ terms}} \underbrace{|\mathrm{trcl}(\{f(x)\})|}_{\text{each } < \kappa} \\ &< \kappa \end{aligned}$$

There are strictly fewer than κ terms because $A \subseteq \mathrm{trcl}(\{A\})$, and a sum of fewer than κ cardinals, each below κ, is below κ because κ is regular. So $B \in H_\kappa$.

This is exactly what Replacement in H_κ needs. If $A, \bar{z} \in H_\kappa$ and each $x \in A$ has a unique $y \in H_\kappa$ with $\varphi^{H_\kappa}(x, y, \bar{z})$, let $f(x)$ be that y; f is a set by Replacement in V, as $A \subseteq H_\kappa$, and $B = \mathrm{range}(f) \in H_\kappa$ is the witness the axiom asks for. □

Power Set, on the other hand, fails at every successor cardinal.

Lemma 3.7.14. *If $\lambda = \kappa^+$, then*

$$(\neg \textit{(Power Set)})^{H_\lambda}$$

Proof. $\mathcal{P}(\kappa) \subseteq H_\lambda$ but $\mathcal{P}(\kappa) \notin H_\lambda$. □

In fact, we can say exactly when it holds.

Lemma 3.7.15. *TFAE:*

(1) $(\textit{Power Set})^{H_\lambda}$

(2) *λ is a strong limit*

Proof.

(2) → (1). Assume (2). Let $A \in H_\lambda$. Then

$$|\mathrm{trcl}(\{\mathcal{P}(A)\})| = |\{\mathcal{P}(A)\} \cup \mathcal{P}(A) \cup \mathrm{trcl}(A)| \leq 1 \oplus 2^{|A|} \oplus |\mathrm{trcl}(A)| < \lambda$$

¬(2) → ¬(1). We show, effectively, that if $\kappa < \lambda$ is a cardinal with $2^\kappa \geq \lambda$, then $\kappa \in H_\lambda$ and $\mathcal{P}(\kappa) \subseteq H_\lambda$ but $\mathcal{P}(\kappa) \notin H_\lambda$, so Power Set fails in H_λ.

□

Putting all this together, we get the following.

Corollary 3.7.16.

1. $H_\omega = V_\omega$ *models* **ZFC − I**
2. $H_{\omega_1} = H_{\omega_1}$ *and all other* $H_{\omega_2}, H_{\omega_3}, \ldots$ *model* **ZFC − P**

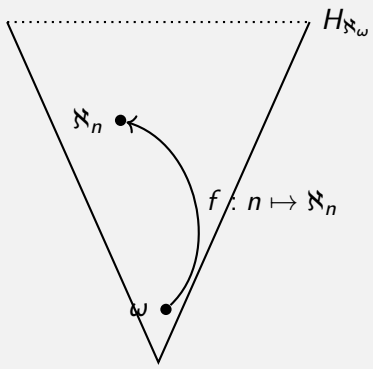
The first cardinal not covered by the corollary is $\aleph_\omega$, which is singular, and it shows that the regularity hypothesis in Lemma 3.7.13 cannot be dropped.

Example 3.7.17 ($H_{\aleph_\omega}$). Everything but Replacement and Power Set holds in $H_{\aleph_\omega}$: Extensionality, Comprehension, Pairing, Union, Choice and Foundation hold in every H_κ, and Infinity holds because $\aleph_\omega > \omega$.

Replacement fails. There's a formula $\varphi(u, v)$ so that, for all $n \in \omega$ and $\kappa \in H_{\aleph_\omega}$,

$$\varphi(n, \kappa)^{H_{\aleph_\omega}} \leftrightarrow \kappa = \aleph_n$$

Namely, $\varphi(n, \kappa)$ says that there is a function g with domain $n+1$ such that $g(0) = \omega$, each $g(i+1)$ is the least cardinal above $g(i)$, and $g(n) = \kappa$. This relativizes correctly because the witness $\langle \aleph_i \mid i \leq n \rangle$ lies in $H_{\aleph_\omega}$, and because an ordinal below $\aleph_\omega$ is a cardinal in $H_{\aleph_\omega}$ iff it is one in V, as every bijection between two ordinals below $\aleph_\omega$ lies in $H_{\aleph_\omega}$. So φ defines $f : n \mapsto \aleph_n$ over $H_{\aleph_\omega}$, on the domain $\omega \in H_{\aleph_\omega}$, but $f \notin H_{\aleph_\omega}$.



Indeed, any B with every $\aleph_n \in B$ has $\aleph_\omega = \bigcup_{n<\omega} \aleph_n \subseteq \mathrm{trcl}(\{B\})$, so $|\mathrm{trcl}(\{B\})| \geq \aleph_\omega$ and $B \notin H_{\aleph_\omega}$. So no set in $H_{\aleph_\omega}$ collects the values of f.

Whether Power Set holds in $H_{\aleph_\omega}$ is, by Lemma 3.7.15, the question of whether $\aleph_\omega$ is a strong limit, which **ZFC** does not settle.

3.7.3 Σ_1-Correctness

The most useful property of the H_κ is that they are right about every Σ_1 statement, and proving it uses the satisfaction relation of Section 3.6.

Theorem 3.7.18. *If κ is an uncountable cardinal, then $H_\kappa \prec_{\Sigma_1} V$.*

Remark. This means that for each Σ_1 formula $\varphi(\bar{u})$ and each matching $\bar{x} \in H_\kappa$, $\varphi(\bar{x})^{H_\kappa} \leftrightarrow \varphi(\bar{x})^V$. In other words, it means Σ_1-formulae are absolute for H_κ. We also say, "H_κ is Σ_1-correct in V."

Note that $H_\kappa \prec_{\Sigma_1} V$ means Σ_1 formulas are absolute between H_κ and V. We borrow model theoretic notation for being an elementary substructure, but note that this is the model theory we know in the "outside world", not the model theory we have inside of V.

Proof. We already know upwards absoluteness, so the only remaining question is downwards absoluteness.

Suppose $\varphi(u)$ is the formula $\exists v\,(\psi(u, v))$, with ψ Δ_0. Suppose $x \in H_\kappa$ is so that $\varphi(x)$.

We need to show that

$$(\exists v\,(\psi(x, v)))^{H_\kappa}$$

Pick any y such that $\psi(x, y)$. Let $N = \mathrm{trcl}(\{x, y\})$. Then $x, y \in N$ and N is transitive. By Δ_0 absoluteness, we know $\psi(x, y)^N$. Can write "$N \vDash \psi(x, y)$".

Apply Downwards Löwenheim-Skolem to find $Z \prec N$ with $\mathrm{trcl}(\{x\}) \subseteq Z$ and $|Z| \leq |\mathrm{trcl}(\{x\})| + \aleph_0$. As $x \in H_\kappa$ and κ is uncountable, $|Z| < \kappa$.

N is transitive, so Extensionality holds in N, and hence in $Z \prec N$; so $\in$ is extensional on Z. It is also well-founded and set-like there, so by the Mostowski Collapse Theorem Scheme (Theorem 2.2.4) there is a transitive set M and an isomorphism $\pi : (Z, \in) \to (M, \in)$, given by $\pi(z) = \{\pi(w) \mid w \in z \cap Z\}$. Two things about π:

- $\pi(w) = w$ for every $w \in \mathrm{trcl}(\{x\})$. Indeed, by $\in$-induction: such a w is a subset of the transitive set $\mathrm{trcl}(\{x\}) \subseteq Z$, so $w \cap Z = w$, and $\pi(w) = \{\pi(v) \mid v \in w\} = w$. In particular, $\pi(x) = x$.
- $|M| = |Z| < \kappa$.

Now $N \vDash \exists v\, \psi(x, v)$, so, as $Z \prec N$ and $x \in Z$, there is $y' \in Z$ with $Z \vDash \psi(x, y')$. Applying the isomorphism, $M \vDash \psi(x, \pi(y'))$, ie, $\psi(x, \pi(y'))^M$. As M is transitive and ψ is Δ_0, Δ_0-absoluteness gives $\psi(x, \pi(y'))$.

Finally, $\pi(y') \in M$ and M is transitive, so $\mathrm{trcl}(\{\pi(y')\}) \subseteq M$ and $|\mathrm{trcl}(\{\pi(y')\})| \leq |M| < \kappa$. So $\pi(y') \in H_\kappa$, and as H_κ is transitive, Δ_0-absoluteness gives $\psi(x, \pi(y'))^{H_\kappa}$. So $(\exists v\, \psi(x, v))^{H_\kappa}$. □

This has a consequence for the levels of the V-hierarchy, through the $\beth$-fixed points.

Recall that $\{\kappa \mid \kappa = \beth_\kappa\}$ is a club class of cardinals. It is closed because the $\beth$ operation is "continuous": if λ is a limit of $\beth$-fixed points, then, as $\alpha \mapsto \beth_\alpha$ is increasing, $\beth_\lambda = \sup_{\alpha<\lambda} \beth_\alpha$ is also the supremum of $\beth_\kappa = \kappa$ over the fixed points $\kappa < \lambda$, which is λ. So a sup of $\beth$-fixed points is a $\beth$-fixed point. It is unbounded because, given any α, the sequence $\kappa_0 = \alpha$, $\kappa_{n+1} = \beth_{\kappa_n}$ is non-decreasing (as $\beth_\gamma \geq \gamma$) and its supremum $\lambda \geq \alpha$ is a fixed point: either the sequence is eventually constant, at a fixed point, or λ is a limit and $\beth_\lambda = \sup_n \beth_{\kappa_n} = \sup_n \kappa_{n+1} = \lambda$.

We also saw that if μ is a strongly inaccessible cardinal, then $\mu = \beth_\mu$ (Proposition 3.5.14).

We have seen that $H_\kappa \prec_{\Sigma_1} V$ for all uncountable cardinals κ. It follows that for such κ, $\kappa = \beth_\kappa$ iff $H_\kappa = V_\kappa$, which in turn implies that $V_\kappa \prec_{\Sigma_1} V$.

Observe that $V_\kappa \prec_{\Sigma_1} V$ is not a formula in the language of set theory. But $\kappa = \beth_\kappa$ and $H_\kappa = V_\kappa$ *are*. By comprehension, put

$$C_1 = \{\kappa \mid \kappa = \beth_\kappa\} = \{\kappa \mid \kappa \text{ is uncountable and } H_\kappa = V_\kappa\}$$

Then C_1 is a class. Writing $C_1 = \{\kappa \mid V_\kappa \prec_{\Sigma_1} V\}$ is problematic.

Note that C_1 is club in **OR**. Indeed, we proved that the set of $\beth$-fixed points contains a club class of ordinals. Define a class function $f : \mathbf{OR} \to \mathbf{OR}$ by

- $f(0) = \beth_0$
- $f(\alpha + 1) = \beth_{f(\alpha)+1}$
- $f(\beta) = \sup_{\alpha<\beta} f(\alpha)$

The range of f is a cub and, for every limit ordinal β, $f(\beta) = \beth_{f(\beta)}$; that is, every element of $\lim(\text{range}(f))$[1] is a $\beth$-fixed point. So C_1 contains a club. So C_1 is unbounded. Moreover, C_1 is closed. So it's a club in **OR**.

3.8 Reflection

Below an inaccessible, we can ask for more than Σ_1-correctness: since V_μ is a set, full elementarity makes sense.

3.8.1 Reflection Below an Inaccessible

Here is the precise statement.

Theorem 3.8.1. *Suppose μ is a strongly inaccessible cardinal. Let $C = \{\kappa < \mu \mid V_\kappa \prec V_\mu\}$. Then C is a club subset of μ.*

[1] This is the class of limit points of range(f), and as we know, the class of limit points of a club is a club

Remark. $V_\kappa \prec V_\mu$ **is** a formula in the language of set theory because both V_κ and V_μ are sets. It would be a different story if they were classes.

"This is model theory. Model theory is math. And math is embedded in set theory. So this makes sense!"

Proof. Clearly C is closed in μ. If $C \cap \lambda \neq \emptyset$ and $\lambda = \sup(C \cap \lambda)$, then we need $\lambda \in C$ (meaning of closed).

For such a λ, we have $V_\lambda = \bigcup_{\kappa<\lambda} V_\kappa = \bigcup_{\kappa \in C \cap \lambda} V_\kappa$.

- It follows, for $\kappa < \kappa' < \mu$ with $\kappa, \kappa' \in C$, that $V_\kappa \prec V_{\kappa'}$.
- Then, $\langle V_\kappa \mid \kappa \in C \cap \lambda \rangle$ is a chain of elementary substructures of V_μ, so its union is also an elementary substructure of V_μ.
- $V_\lambda \prec V_\mu$, so $\lambda \in C$.
- It also follows that $V_\kappa \prec V_\lambda$ for $\kappa \in C \cap \lambda$.

Now let $\kappa_0 < \mu$ be an arbitrary uncountable cardinal. By Downwards Löwenheim-Skolem, we have $X_0 \prec V_\mu$ with $V_{\kappa_0} \cup \{\kappa_0\} \subseteq X_0$ and $|X_0| = |V_{\kappa_0} \cup \{\kappa_0\}| = \beth_{\kappa_0} < \beth_\mu = \mu$.

`sorry` □

3.8.2 Hierarchies

To state reflection in general, we give a name to the shape that $\langle V_\alpha \mid \alpha \in \mathbf{OR} \rangle$ and the H_κ share.

Definition 3.8.2 (Hierarchy). We say that a class function $\langle Z_\alpha \mid \alpha \in C \rangle$ is a **hierarchy** of sets iff

- each Z_α is a transitive set
- C is a club class of ordinals
- for every $\alpha < \beta$ from C, $Z_\alpha \subset Z_\beta$

- for every β a limit point[a] of C,

$$Z_\beta = \bigcup_{\alpha \in C \cap \beta} Z_\alpha$$

We usually write

$$Z = \bigcup_{\alpha \in C} Z_\alpha = \{x \mid \exists \alpha \in C \text{ s.t. } x \in Z_\alpha\}$$

(unless, of course, we already have a name for our hierarchy).

[a] as C is club, such βs do also lie in C

We have already seen two of these.

Example 3.8.3. The following are all examples of hierarchies we've seen:

$$\langle V_\alpha \mid \alpha \in \mathbf{OR} \rangle \quad , \quad \langle H_\kappa \mid \kappa \text{ an infinite cardinal} \rangle$$

Indeed, there will be examples where $Z \neq V$.

3.8.3 The Reflection Theorem

Every hierarchy reflects any finitely many formulae down to club many of its levels.

Theorem 3.8.4 (Reflection theorem scheme, proved in **ZF**). *Suppose φ_i for $i < \ell < \omega$ are formulae in the language of set theory. Suppose that $\langle Z_\alpha \mid \alpha \in C \rangle$ is a hierarchy. Then there's a club class $D \subseteq C$ such that for every $\alpha \in D$ and $i < \ell$, φ_i is absolute between Z_α and Z.*

Proof. WLOG, assume that for all $j < \ell$ and proper subformulae ψ of φ_j, there is some $i < j$ such that ψ is φ_i. Ie, we are setting up an induction on the complexity of φ by setting up an induction on ordinals.

Let's define a class sequence of ordinals $\langle \alpha_\eta \mid \eta \in \mathbf{OR} \rangle$ as follows.

- Base: Define $\alpha_0 = \min(C)$.

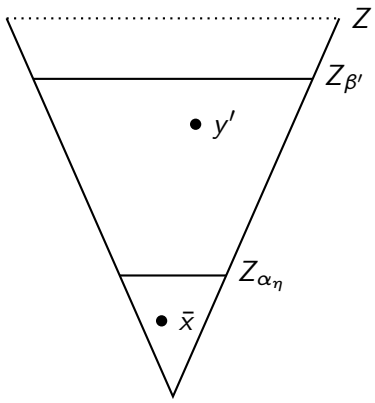
- Successor: Given α_η, define $\alpha_{\eta+1}$ to be the least $\beta \in C \setminus (\alpha_\eta + 1)$ so that for all $j < \ell$, if $\varphi_j(\bar{u})$ is of the form $\exists v, \varphi_i(\bar{u}, v)$, then for all $\bar{x}$ from Z_{α_η} that match $\bar{u}$, if there is $y \in Z$ such that $\varphi_i(\bar{x}, y)^Z$, then there is $y \in Z_\beta$ such that $\varphi_i(\bar{x}, y)^Z$.

- Limit case: If θ is a limit ordinal, define

$$\alpha_\theta = \sup_{\eta<\theta}(\alpha_\eta) \in \lim(C) \subset C$$

(because C is club).

The successor step is where the work happens. A tuple $\bar{x}$ from Z_{α_η} may have a witness y' only higher up, in some $Z_{\beta'}$, and $\alpha_{\eta+1}$ is chosen high enough to catch a witness for every such $\bar{x}$ at once:



Only existential formulae need this treatment. We may take every φ_j to be built from atomic formulae using $\neg$, $\wedge$ and $\exists$ alone, writing $\forall$ as $\neg\exists\neg$. Relativization leaves atomic formulae unchanged, so they mean the same in Z_α as in Z, and it commutes with $\neg$ and $\wedge$, so absoluteness passes through them. The one place it can fail is $\exists v$, where Z may have a witness that Z_α lacks.

The successor step makes sense. For each $j < \ell$ of the form above and each $\bar{x}$ from Z_{α_η} that has a witness $y \in Z$, there is a least $\gamma \in C$ with a witness in Z_γ, as $Z = \bigcup_{\gamma \in C} Z_\gamma$. There are only set many such pairs $(j, \bar{x})$, as Z_{α_η} is a set, so these γ have a supremum, and as C is unbounded there is some $\beta \in C$ above it and above α_η. Any such β works, since $Z_\gamma \subseteq Z_\beta$ for $\gamma \leq \beta$ in C.

Then take D to be $\{\alpha_\theta \mid \theta \text{ is a limit ordinal}\}$. The sequence $\langle \alpha_\eta \mid \eta \in \mathbf{OR} \rangle$ is strictly increasing and continuous at limits, so D is a club class: it is unbounded since $\alpha_\theta \geq \theta$, and closed since a supremum of α_θs over limit ordinals θ in some set Θ with no largest element is $\alpha_{\sup\Theta}$, and $\sup\Theta$

is a limit ordinal. And $D \subseteq C$ by the limit case.

Finally, fix $\alpha = \alpha_\theta \in D$; we show that each φ_i is absolute between Z_α and Z, by induction on i. The atomic and propositional cases are as above, so suppose $\varphi_j(\bar{u})$ is $\exists v\, \varphi_i(\bar{u}, v)$ with $i < j$, and let $\bar{x}$ from Z_α match $\bar{u}$.

From Z_α to Z. If $y \in Z_\alpha$ and $\varphi_i(\bar{x}, y)^{Z_\alpha}$, then $y \in Z$ and $\varphi_i(\bar{x}, y)^{Z}$ by the induction hypothesis, so $\varphi_j(\bar{x})^{Z}$.

From Z to Z_α. Suppose $\varphi_j(\bar{x})^{Z}$. As α_θ is a limit point of C, $Z_{\alpha_\theta} = \bigcup_{\gamma \in C \cap \alpha_\theta} Z_\gamma = \bigcup_{\eta < \theta} Z_{\alpha_\eta}$, and $\bar{x}$ is a finite tuple, so $\bar{x}$ is from Z_{α_η} for some $\eta < \theta$. By the choice of $\alpha_{\eta+1}$, there is $y \in Z_{\alpha_{\eta+1}} \subseteq Z_\alpha$ with $\varphi_i(\bar{x}, y)^{Z}$, and by the induction hypothesis $\varphi_i(\bar{x}, y)^{Z_\alpha}$. So $\varphi_j(\bar{x})^{Z_\alpha}$.

□

Here's an interesting corollary.

Let $\sigma_0, \ldots, \sigma_\ell$ be finitely many axioms of **ZFC**. By the reflection theorem scheme, there are club many $\alpha \in \mathbf{OR}$ so that $\forall i \leq \ell$, $\sigma_i^{V_\alpha}$. Equivalently, $\forall i \leq \ell$, $V_\alpha \vDash \sigma_i$. Rephrasing:

Corollary 3.8.5. *For every finite sub-theory T of* **ZFC**,

$$\mathbf{ZFC} \vdash \text{“}T \text{ has a transitive model”}$$

where the quotes (“read with a different tone of voice”)

With Choice, we can make the model countable as well. Given α with

$$V_\alpha \vDash \bigwedge_{i \leq \ell} \sigma_i$$

we can apply Downwards Löwenheim-Skolem to get some countable $X \prec V_\alpha$, so that $X \vDash \bigwedge_{i \leq \ell} \sigma_i$ too. As V_α is transitive, it satisfies Extensionality, and hence so does X; so $\in$ is extensional on X, as well as well-founded and set-like. Then apply Mostowski (Theorem 2.2.4) to get $X \simeq M$ with M transitive and countable, and $M \vDash \bigwedge_{i \leq \ell} \sigma_i$, because satisfaction is preserved by isomorphisms. So in fact **ZFC** proves that every finite sub-theory of **ZFC** has a countable transitive model.

Chapter 4

Gödel's Constructible Universe

We previewed Gödel's constructible universe at the very start of Chapter 1, where we defined the operation $\mathcal{D}$ informally and promised to explore it in great detail. We now build it properly, using everything Chapter 3 has given us: satisfaction to define it, absoluteness to compute it inside transitive models, and reflection to check the axioms in it.

4.1 Definable Subsets

The universe is built by iterating one operation, which takes the subsets of a set that the set itself can define.

Definition 4.1.1 (Definable Subsets). Define the class function $A \mapsto \mathcal{D}(A)$ by

$$\mathcal{D}(A) = \{X \subseteq A \mid X \text{ is definable with parameters from } A \text{ over the structure } (A, \in)\}$$

Ie, $X \in \mathcal{D}(A)$ iff there are $\bar{a} \in A^{<\omega}$ and $n < \omega$ such that, letting φ be the nth formula, we have that

$$X = \{b \in A \mid (A, \in) \vDash \varphi(b, \bar{a})\}$$

Recall that $(A, \in) \vDash \varphi(b, \bar{a})$ can be written $A \vDash \varphi(b, \bar{a})$, and that this holds iff $\varphi(b, \bar{a})^A$ (Lemma 3.6.1).

We don't need the Power Set Axiom to know that $\mathcal{D}(A)$ exists for all A, ie, $\mathbf{ZF} - \mathbf{P}$ is enough.

Use Replacement to know that $\mathcal{D}(A)$ is the range of the class function

$$(\bar{a}, n) \mapsto \text{the subset of } A \text{ defined by the } n\text{th formula and parameters } \bar{a}$$

Its domain is contained in $A^{<\omega} \times \omega$, so it's a set. By Replacement, we know the range is a set.

That is the first half of the following lemma, and the second half is what will let us compute $\mathcal{D}$ inside transitive models.

Lemma 4.1.2 (Properties of the Class Function $A \mapsto \mathcal{D}(A)$).

1. $A \mapsto \mathcal{D}(A)$ *is defined in* $\mathbf{ZF} - \mathbf{P}$.
2. $A \mapsto \mathcal{D}(A)$ *is absolute for transitive class models of* $\mathbf{ZF} - \mathbf{P}$.

Proof sketch. We defined formulae to be certain members of **HF**: "φ is a formula" is absolute.

We defined $A \vDash \varphi(\bar{a})$, for a formula φ and $\bar{a} \in A^{<\omega}$, by recursion on the complexity of φ. This is absolute.

Also note that $A \vDash \exists a_1 \forall a_2 \exists a_3 \cdots Q a_n \psi$, with ψ quantifier-free, is equivalent to the Δ_0 formula

$$\exists a_1 \in A \forall a_2 \in A \exists a_3 \in A \cdots Q a_n \in A \psi$$

The absoluteness of the recursive definition says more than this: it is the class function $(A, \varphi, \bar{a}) \mapsto \{0, 1\}$ that is absolute. □

On finite sets, $\mathcal{D}$ is nothing new.

Lemma 4.1.3. *If A is finite, then $\mathcal{D}(A) = \mathcal{P}(A)$.*

Proof sketch. Just unfold the definitions. □

4.2 The L-Hierarchy

Now we iterate $\mathcal{D}$ through the ordinals, exactly as we iterated $\mathcal{P}$ to build the V-hierarchy.

Definition 4.2.1 (The L-Hierarchy). Define $L_0 = 0$, $L_{\alpha+1} = \mathcal{D}(L_\alpha)$ for all ordinals α, and for all limit ordinals β, $L_\beta = \bigcup_{\alpha<\beta} L_\alpha$. Then define the universe $L = \bigcup_{\alpha\in\mathbf{OR}} L_\alpha$.

One big goal (Gödel): working in **ZF**, show that L is a transitive class model of **ZFC** + **GCH**+ lots of other stuff. Also, if M is any transitive proper class model of **ZF**, then $L = L^M \subseteq M$.

Intuition: in the V-hierarchy, we jump from the countable $V_\omega = \mathbf{HF}$ to the uncountable $V_{\omega+1} = \mathcal{P}(V_\omega)$ in one step. Also, assuming **AC** in V, $|V_{\omega+1}| = 2^{\aleph_0}$, and we can't tell how big $2^{\aleph_0}$ is!

We'll see that $L_\omega = \mathbf{HF} = V_\omega$, but $L_{\omega+1} = \mathcal{D}(L_\omega)$ is also countable. Why? Count formulae and parameters from L_ω. By using $\mathcal{D}$ instead of $\mathcal{P}$, we slow things way down: there's hope of understanding and answering questions.

We can show that $\langle L_\alpha \mid \alpha \in \mathbf{OR}\rangle$ is a hierarchy of sets (Definition 3.8.2), ie, $L_\alpha \subset L_\beta$ for $\alpha < \beta$, and each L_α is a transitive set. We can also check that $L_\alpha \cap \mathbf{OR} = \alpha$.

Since $\mathcal{D}$ agrees with $\mathcal{P}$ on finite sets, the two hierarchies agree for as long as their levels stay finite, and a little beyond.

Corollary 4.2.2. *For all $n \leq \omega$, $L_n = V_n$.*

Proof sketch. Consider $n = 0$, n successor and $n = \omega$, in all cases we're ok. □

Let's now talk briefly about cardinality. Note that we will not assume choice here, so in particular, we cannot conclude every set has a cardinality. So we need to think about things in terms of well-orderings.

Lemma 4.2.3. *If A has a cardinality and it's infinite, then $\mathcal{D}(A)$ has a cardinality and $|\mathcal{D}(A)| = |A|$.*

Proof. Say $|A| = \alpha$. Notice that

$$|\mathcal{D}(A)| \leq \left|\underbrace{\text{Formulae}}_{\text{countable}} \times A^{<\omega}\right| = |A|$$

□

Iterating this through the L-hierarchy gives the size of every level.

Corollary 4.2.4. *Each L_α has a cardinality and, for $\alpha \geq \omega$, $|L_\alpha| = |\alpha|$.*

Proof sketch. Argue by recursion on $\alpha \in \mathbf{OR}$. Base and successor cases are not hard. For the limit case, just end-extend all well-orderings of L_α for $\alpha < \beta$, β being some limit ordinal. This gives a well-ordering. □

With this we end our discussion of cardinality.

Finally, the L-hierarchy never outgrows the V-hierarchy.

Lemma 4.2.5. *For all $\alpha \in \mathbf{OR}$, $L_\alpha \subseteq V_\alpha$.*

Proof sketch. $\mathcal{D}(A) \subseteq \mathcal{P}(A)$. □

4.3 The Axioms in L

We now come to the most important theorem about L. We work in **ZF**.

Theorem 4.3.1 (Gödel). *For each axiom σ of **ZFC** $\cup$ **GCH**, we have σ^L.*

Proof.

- Extensionality. L is transitive.
- Pairing. Let $x, y \in L$. Pick $\alpha \in \mathbf{OR}$ such that $x, y \in L_\alpha$. Then $\{x, y\} \in \mathcal{D}(L_\alpha) = L_{\alpha+1} \subset L$.
- Comprehension. Let $A \in L$, $\bar{y} \in L$, $B = \left\{x \in A \mid \varphi^L(x, \bar{y})\right\}$. We need $B \in L$.

  Apply the Reflection Theorem Scheme (Theorem 3.8.4) to the L-hierarchy to find α such that φ is absolute between L_α and L. We can do this with $A \in L_\alpha$ and $\bar{y} \in L_\alpha$.

Then, since $A \subseteq L_\alpha$,

$$B = \{x \in L_\alpha \mid L_\alpha \vDash (x \in A \wedge \varphi(x, \bar{y}))\} \in \mathcal{D}(L_\alpha) = L_{\alpha+1} \subset L$$

- Unions. Let $\mathcal{F} \in L$. Pick α such that $\mathcal{F} \in L_\alpha$. Then,

  $$\bigcup \mathcal{F} = \{x \in L_\alpha \mid L_\alpha \vDash \exists y\ (y \in \mathcal{F} \wedge x \in y)\} \in \mathcal{D}(L_\alpha) = L_{\alpha+1} \subset L$$

  Indeed, L_α is transitive, so every $y \in \mathcal{F}$ lies in L_α, and then so does every $x \in y$. So $\bigcup \mathcal{F} \subseteq L_\alpha$, and for $x \in L_\alpha$, $L_\alpha \vDash \exists y\ (y \in \mathcal{F} \wedge x \in y)$ iff some $y \in \mathcal{F}$ has $x \in y$. In particular $\bigcup \mathcal{F}$ is a member of L containing every member of $\mathcal{F}$ as a subset, which is what the Axiom of Unions asks of L.

- Power Set. Let $A \in L$. We need $\mathcal{P}(A) \cap L \in L$. It is enough to show that there is some $B \in L$ with $\mathcal{P}(A) \cap L \subseteq B$.

  In V, consider the class function $X \mapsto$ the least ξ such that $X \in L_\xi$, with domain $\mathcal{P}(A) \cap L$. By Replacement in V (plus stuff), we know there's some α such that $\forall X \in \mathcal{P}(A) \cap L\ (X \in L_\alpha)$. Take $B = L_\alpha \in L$; this works.

- Replacement. Assume $A, \bar{z} \in L$ and $\forall x \in A\, \exists! y \in L\, \varphi^L(x, y, \bar{z})$. Pick α such that $A, \bar{z} \in L_\alpha$. Consider the class function

  $$x \mapsto \text{the least } \xi \text{ such that } \exists y \in L_\xi\, \varphi^L(x, y, \bar{z})$$

  By Replacement in V, there's some β such that all such ξs are $< \beta$. Then $L_\beta \in L$ and $\forall x \in A\, \exists y \in L_\beta\, \varphi^L(x, y, \bar{z})$.

- Foundation. $L \subseteq V = \mathbf{WF}$.

- Infinity. $\omega \in L$.

That covers every axiom of **ZF**. Choice and **GCH** take more work: they are Theorem 4.4.1 and Theorem 4.5.1 below. □

So now we know L models **ZF**. By the absoluteness of $A \mapsto \mathcal{D}(A)$ and $\alpha \mapsto L_\alpha$ (Lemma 4.1.2 and

Section 3.4.3), $L^L = L$, and $L^M = L_{\mathbf{OR}\cap M}$ for every transitive class model M of $\mathbf{ZF} - \mathbf{P}$: either $\mathbf{OR} \cap M = \mathbf{OR}$ and $L^M = L$, or $\mathbf{OR} \cap M = \alpha \in \mathbf{OR}$ and $L^M = L_\alpha$. In fact, there is a finite subtheory of $\mathbf{ZF} - \mathbf{P}$ that's enough for this.

In particular, if M is a transitive proper class model of $\mathbf{ZF}$, then $\mathbf{OR} \cap M = \mathbf{OR}$ (Section 3.4.4), so $L = L^M \subseteq M$: this is the second half of Gödel's goal.

4.4 A Well-Ordering of L

In the proof of Corollary 4.2.4, we indicated how to define a well-ordering $<_\alpha$ of L_α by recursion on $\alpha \in \mathbf{OR}$. Here it is in full.

- $<_0 = \emptyset$
- For $x, y \in L_{\alpha+1} = \mathcal{D}(L_\alpha)$, put $x <_{\alpha+1} y$ iff one of the following occurs:
  1. $x, y \in L_\alpha$ and $x <_\alpha y$
  2. $x \in L_\alpha$ and $y \in L_{\alpha+1} \setminus L_\alpha$
  3. $x, y \in L_{\alpha+1} \setminus L_\alpha$ and $(k, p) <_{\text{lex}} (l, q)$, where (k, p) is the lexicographically least pair in $\omega \times L_\alpha^{<\omega}$ such that if φ is the kth formula, then $x = \{z \in L_\alpha \mid L_\alpha \vDash \varphi(z, p)\}$, and (l, q) is the analogous thing for y

  Note that $<_{\alpha+1}$ end-extends $<_\alpha$.
- For limit ordinals β, put

$$<_\beta = \bigcup_{\alpha<\beta} <_\alpha$$

This works: $\alpha \mapsto <_\alpha$ is definable without parameters; moreover, it is absolute for transitive models of a certain finite subtheory of $\mathbf{ZF} - \mathbf{P}$. We fix a name for it.

Notation. For the rest of this chapter, T will denote a finite subtheory of $\mathbf{ZF} - \mathbf{P}$ such that $\alpha \mapsto L_\alpha$ and $\alpha \mapsto <_\alpha$ are both absolute for transitive models of T.

Notice that if we put

$$<_L = \bigcup_{\alpha \in \mathbf{OR}} <_\alpha$$

then $<_L$ is a class well-ordering of L. Note that it is set-like, extensional and well-founded, so it admits a Mostowski Collapse. This Mostowski Collapse turns out to simply be **OR**.

For any transitive class model M of T,

$$(<_L)^M = \begin{cases} <_L & \text{if } \mathbf{OR} \cap M = \mathbf{OR} \\ <_\alpha & \text{if } \mathbf{OR} \cap M = \alpha \end{cases}$$

The first thing this buys us is Choice in L.

Theorem 4.4.1 (using **ZF** in V). $(\mathbf{AC})^L$

Proof. Let $A \in L$. Pick α such that $A \in L_\alpha$. Then $A \subseteq L_\alpha$. Let $R = <_\alpha \cap (A \times A)$. Since $(<_\alpha)^L = <_\alpha$, we know $R \in L$. We know (A, R) is a well-ordering. This is Π_1, hence downward absolute to L. □

The well-ordering $<_L$ also tells us that every constructible set can be defined from an ordinal. To say this, we need the classes of sets definable from ordinals. We cannot just say "x is definable from finitely many ordinals", as there is no formula expressing definability (cf. Section 3.6); so we ask for a definition over some V_α instead.

Definition 4.4.2 (Ordinal Definability). A set x is **ordinal definable** iff there are an ordinal α with $x \in V_\alpha$, a formula $\varphi(u, \bar{v})$ and ordinals $\bar{\xi} \in \alpha^{<\omega}$ such that x is the unique $y \in V_\alpha$ with $V_\alpha \vDash \varphi(y, \bar{\xi})$. We write **OD** for the class of ordinal definable sets.
A set x is **hereditarily ordinal definable** iff $\mathrm{trcl}(\{x\}) \subseteq \mathbf{OD}$. We write **HOD** for the class of hereditarily ordinal definable sets.

By the Reflection Theorem Scheme, this is what we wanted: for each formula $\varphi(u, \bar{v})$, if $\bar{\xi}$ are ordinals and x is the unique set with $\varphi(x, \bar{\xi})$, then reflecting φ to a level V_α with $x, \bar{\xi} \in V_\alpha$ shows $x \in \mathbf{OD}$. Conversely, a set in **OD** is defined in V from the ordinals α and $\bar{\xi}$ by the formula "$u \in V_\alpha$

and $V_\alpha \vDash \varphi(u, \bar{\xi})$".

So L sits inside **HOD**, and L knows it.

Theorem 4.4.3. *$L \subseteq$ **HOD**, and $\mathbf{HOD}^L = L$.*

Proof. Note that given any $x \in L$, we have some $\xi \in \mathbf{OR}$ such that x is the ξth member of L according to $<_L$. In other words, if $G : (L, <_L) \simeq \mathbf{OR}$ then $G(x) = \xi$. Thus, x is definable from ξ, so $x \in \mathbf{OD}$. Since L is transitive, $x \in \mathbf{HOD}$. This calculation is absolute down to L. So $\mathbf{HOD}^L = L$. □

4.5 The Generalized Continuum Hypothesis in L

The last axiom on Gödel's list takes the most work, and its proof uses an idea particular to L.

Theorem 4.5.1 (Working in L). $(\mathbf{GCH})^L$. *Ie,*

$$\big(\textit{For all infinite cardinals } \kappa,\ \kappa^+ = 2^\kappa\big)^L$$

Proof, working in L and keeping in mind that $L^L = L$. Let λ be an infinite cardinal. We will show that $H_\lambda = L_\lambda$. This will be enough because if we have any infinite cardinal κ, setting $\lambda = \kappa^+$, we get that

$$\mathcal{P}(\kappa) \subseteq H_\lambda = L_\lambda$$

meaning that

$$2^\kappa = |\mathcal{P}(\kappa)| \leq |L_\lambda| = |\lambda| = \lambda = \kappa^+$$

And of course the other inequality is always true in L, as in any model of **ZFC**.

So now let's prove that $H_\lambda = L_\lambda$ for all infinite cardinals λ.

Fix an infinite cardinal λ. We already know that

$$H_\omega = V_\omega = L_\omega$$

so we may assume that λ is uncountable.

- $\underline{L_\lambda \subseteq H_\lambda}$. Let $z \in L_\lambda$. Pick α with $\omega \leq \alpha < \lambda$ such that $z \in L_\alpha$. Then certainly $\mathrm{trcl}(\{z\}) \subseteq L_\alpha$. So

$$|\mathrm{trcl}(\{z\})| \leq |L_\alpha| = |\alpha| \leq \alpha < \lambda$$

  So $z \in H_\lambda$.

- $\underline{H_\lambda \subseteq L_\lambda}$. Let $z \in H_\lambda$. Recall our finite subtheory T of $\mathbf{ZF} - \mathbf{P}$, for whose transitive models $\alpha \mapsto L_\alpha$ and $\alpha \mapsto <_\alpha$ are absolute, maybe augmented with $V = L$. By the reflection theorem scheme, there's some β such that $z \in L_\beta$ and σ^{L_β} for all $\sigma \in T$ and $(V = L)^{L_\beta}$. So $L_\beta \vDash T$ and $(V = L)^{L_\beta}$.

  Observe: $\mathrm{trcl}(\{z\}) \subseteq L_\beta$ and $|\mathrm{trcl}(\{z\})| < \lambda$. By Downwards Löwenheim-Skolem, we have some $X \preceq L_\beta$ with $|X| < \lambda$ and $\mathrm{trcl}(\{z\}) \subseteq X$. Let $G : X \simeq M$ be the Mostowski Collapse of X, so M is a transitive model of T. Since $\mathrm{trcl}(\{z\}) \subseteq X$, we know that

$$G \restriction \mathrm{trcl}(\{z\}) = \mathrm{id} \restriction \mathrm{trcl}(\{z\})$$

  In particular, $G(z) = z \in M$.

  As $L_\beta \vDash V = L$, we also have $M \vDash V = L$. In other words, $(V = L)^M$. This says that $V^M = L^M$, which says $M = L_\alpha$ where $\alpha = \mathbf{OR} \cap M$. Therefore, $z \in L_\alpha$ and $|X| = |M| = |L_\alpha| = |\alpha|$, and this is less than λ, so $\alpha < \lambda$.

□

We do something very interesting in the above proof: we take some nice level of the L hierarchy (which we get by the reflection theorem scheme), we take an elementary substructure (which we get by downwards Löwenheim-Skolem), then take its Mostowski collapse, and we end up with *another* level of the L hierarchy.

It turns out that this is not true of the V- or the H-hierarchy. The phenomenon has a name.

Definition 4.5.2 (Condensation). The phenomenon that the Mostowski collapse of an elementary substructure of a suitable level of the L-hierarchy—such as the L_β in the proof above, with $L_\beta \vDash T$ and $(V = L)^{L_\beta}$—is again a level of the L-hierarchy is known as **condensation**.

The condensation used in that proof can be stated once and for all, for any level of the L-hierarchy satisfying enough of $\mathbf{ZF} - \mathbf{P}$.

Lemma 4.5.3 (Condensation Lemma). *Let T be some finite subtheory of $\mathbf{ZF} - \mathbf{P}$ such that for transitive class models of T,*

$$A \mapsto \mathcal{D}(A) \qquad \alpha \mapsto L_\alpha \qquad \alpha \mapsto <_\alpha \qquad L \qquad <_L$$

are all absolute. By the reflection theorem (scheme), $\{\beta \mid L_\beta \vDash T\}$ is unbounded in $\mathbf{OR}$. Given $L_\beta \vDash T$ and $X \preceq L_\beta$, if $M \simeq X$ and M is transitive, then $M = L_\alpha$, where $\alpha = \mathbf{OR} \cap M$.

Here's another way to view the condensation lemma.

Given a transitive M and an elementary embedding $\pi : M \hookrightarrow L_\beta$ (for some β as above), we conclude that there is some $\alpha \leq \beta$ such that $M = L_\alpha$. In the statement above, $X = \mathrm{ran}(() \pi)$, with π being the inverse of the Mostowski collapse of X.

4.6 A Δ^1_2 Well-Ordering of the Reals

Before proceeding further, let's note that working in ZFC and proving φ^L is tantamount to working in ZFC with the further assumption that $V = L$ (ie, assuming $\forall x \exists \alpha \, (x \in L_\alpha)$) and proving φ. The reason for this is that $L^L = L$. Throughout this subsection, we assume $V = L$.

Let's now ask ourselves the following question: how complex is $<_L$ restricted to ${}^\omega\omega$? Denote $<_{\omega\omega} = <_L \cap ({}^\omega\omega \times {}^\omega\omega)$.

Working in L, we know that for every $x, y \in {}^\omega\omega$, $x <_L y$ iff there is a countable, transitive M with $x, y \in M$ such that

$$M \vDash T \cup \{V = L, x <_L y\}$$

This is in turn equivalent to

$$\mathbf{HC} \vDash \exists M \text{ transitive, countable s.t. } x, y \in M \text{ and } M \vDash T \cup \{V = L, x <_L y\}$$

This shows that $<_{\omega_\omega}$ is Σ_1 definable over $\mathbf{HC}$ without parameters. It will follow from a homework exercise that $<_{\omega_\omega}$ is Σ^1_2 (lightface).

Note that $x \not<_L y$ iff $\exists M$ countable and transitive with $x, y \in M$ and $M \vDash T \cup \{V = L, x \not<_L y\}$. This shows:

- $\not<_{\omega_\omega}$ is $\Sigma_1^{\mathbf{HC}}$
- $<_{\omega_\omega}$ is $\Pi_1^{\mathbf{HC}}$
- $<_{\omega_\omega}$ is $\Delta_1^{\mathbf{HC}}$, thus Δ^1_2

So $<_{\omega_\omega}$ is a Δ^1_2 well-ordering of ${}^\omega\omega$. Note that it has type ω_1: for each y,

$$\{x \mid x <_L y\} \subseteq L_\alpha$$

where $y \in L_\alpha$ for some $\alpha < \omega_1$ (as $y \in H_{\omega_1} = L_{\omega_1}$), so countability of L_α means that $\{x \mid x <_L y\}$ is countable, and the only well-ordering that is uncountable but whose every initial segment is countable is ω_1.

Look at $<_L \cap (\mathbb{R} \times \mathbb{R})$. Ie, view ${}^\omega\omega$ as the actual set of reals. This is also a Δ^1_2 well-ordering of $\mathbb{R}$ of type ω_1, and we will want a shorter name for it.

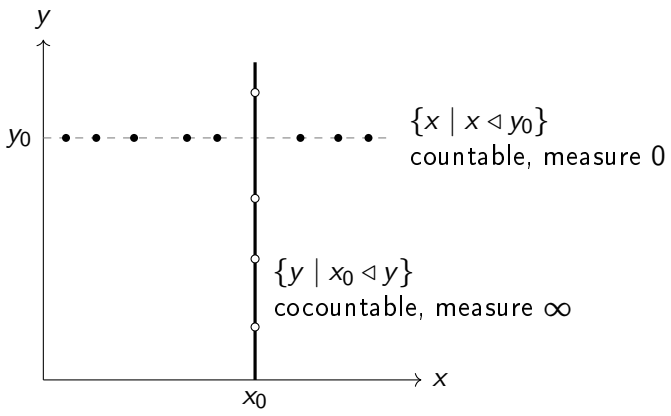
Notation. For the rest of this section, $\lhd$ will denote $<_L \cap (\mathbb{R} \times \mathbb{R})$.

Lemma 4.6.1. *$<_L \cap (\mathbb{R} \times \mathbb{R})$ is a non-Lebesgue measurable subset of $\mathbb{R} \times \mathbb{R}$.*

Proof sketch. For a given y, the set $\{x \mid x <_L y\}$ is countable, therefore has measure zero. So if $\lhd$ is Lebesgue measurable, then it must have Lebesgue measure zero.

However, for a given x, $\{y \mid x <_L y\}$ is cocountable, hence has measure ∞. So if $\lhd$ is Lebesgue measurable, it must have measure ∞.

So $\lhd$ cannot be measurable, because if it is, then these things would both be true, which is impossible! (Note that there is some Fubini going on here because the point is that in both cases, we are getting the measure of the same set, ie, the Lebesgue measure of $\lhd$ *is* the first one (0) and the second one (∞).)



Integrating the measures of the horizontal sections over y gives 0; integrating the measures of the vertical sections over x gives ∞. By Fubini, both would have to be the measure of $\lhd$. □

Chapter 5

Trees and Suslin's Problem

Suslin asked whether the real line is characterized by a condition weaker than separability, and the answer turns out to depend on the universe of sets. This chapter states the question and then develops the combinatorics that bears on it: trees, the tree property, and the principle $\diamondsuit$, from which we construct a Suslin tree.

5.1 Suslin's Problem

We now put L to work on a question about the real line which, on its face, has nothing to do with set theory.

5.1.1 Dense Linear Orderings without Endpoints

Here are some simple facts about dense linear orderings without endpoints. We should be able to prove these using prior knowledge.

We will abbreviate "Dense Linear Ordering without Endpoints" DLO (WE will be implicit).

Lemma 5.1.1. *Up to isomorphism, $\mathbb{Q}$ is the unique countable DLO.*

As a reminder, this is a back-and-forth argument; if we have $\mathbb{Q}$ and another one, then we build an isomorphism by going back and forth.

The reals have a characterization of the same kind.

Lemma 5.1.2. *Up to isomorphism, $\mathbb{R}$ is the unique DLO that is separable (ie, admits a countable, dense subset) and complete (ie, has the least upper-bound property).*

As a reminder, if $(\mathbb{S}, <_{\mathbb{S}})$ is a DLO that's separable and complete, then we get an isomorphism between the dense countable subsets by the previous one, then using the LUB property, we extend that to an isomorphism on the entirety of the two sets.

Suslin's question is about a weakening of separability.

Definition 5.1.3 (Countable Chain Condition). A DLO satisfies the **countable chain condition (CCC)** if every family of pairwise disjoint open intervals is countable.

This is indeed weaker.

Lemma 5.1.4. *Every separable DLO satisfies the CCC.*

5.1.2 Suslin's Question

Suslin asked the following question:

> Is every complete DLO satisfying the CCC isomorphic to $\mathbb{R}$?

Suslin's Hypothesis says "YES". This is often abbreviated **SH**. A counterexample to **SH** is called a Suslin Line. In other words, **SH** says that there are no Suslin lines.

Whether there are any turns out to depend on the universe of sets.

Theorem 5.1.5 (Jensen, 1960s). *If $V = L$, then there exists a Suslin line.*

In other words, $\neg$**SH** holds in L. Since L is a model of **ZFC** in which $V = L$ holds, the following is immediate.

Corollary 5.1.6. *If* **ZFC** *is consistent, then so is* **ZFC** $\cup \{\neg$**SH**$\}$.

The other half cannot come from L, since **SH** fails there.

Theorem 5.1.7 (Solovay–Tennenbaum). *If* **ZFC** *is consistent, then so is* $\mathbf{ZFC} \cup \{\mathbf{SH}\}$.

There is a name for a statement that a theory can neither prove nor refute.

Definition 5.1.8 (Independence). Let Γ be a theory and σ a sentence. We say that σ is **independent of** Γ if $\Gamma \nvdash \sigma$ and $\Gamma \nvdash \neg\sigma$, ie, if $\Gamma \cup \{\sigma\}$ and $\Gamma \cup \{\neg\sigma\}$ are both consistent.

We thus say that **SH** is independent of **ZFC**, as long as **ZFC** is consistent: by the corollary and the theorem above, $\mathbf{ZFC} \cup \{\neg\mathbf{SH}\}$ and $\mathbf{ZFC} \cup \{\mathbf{SH}\}$ are then both consistent.

5.2 Trees

So far, we've only looked at trees on ω or trees on some other set. These are subtrees of $({}^{<\omega}\Omega, \subsetneq)$. Here is the general notion.

Definition 5.2.1 (Tree). $(T, <_T)$ is a **tree** iff it's a strict partial ordering (transitive and irreflexive) and, for all $y \in T$, $\{x \mid x <_T y\}$ is well-ordered by $<_T$.

Fix a tree $(T, <_T)$.

Each node of a tree sits at some height, and the nodes at the same height make up a level.

Definition 5.2.2 (Height and Levels). For $y \in T$, define

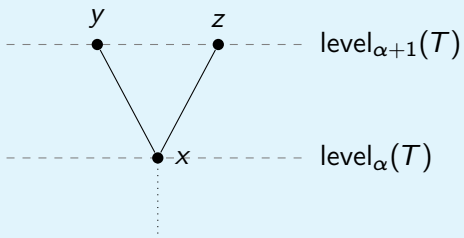
$$\begin{aligned} \text{height}_T(y) &= \text{type}(\{x \mid x <_T y\}, <_T) \\ \text{level}_\alpha(T) &= \{y \in T \mid \text{height}_T(y) = \alpha\} \\ \text{height}(T) &= \text{the least } \alpha \text{ such that } \text{level}_\alpha(T) = \emptyset \end{aligned}$$

We call $\text{height}_T(y)$ the **height** of y, $\text{level}_\alpha(T)$ the **αth level** of T, and $\text{height}(T)$ the **height** of T.

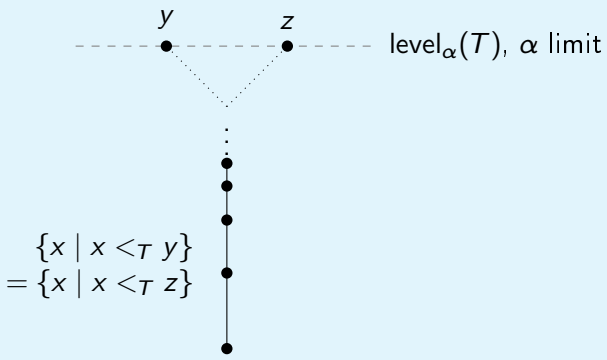
We will mostly care about trees that are as well-behaved as possible.

Definition 5.2.3 (Normal Tree). We say $(T, <_T)$ is **normal** iff

1. T has a unique root
2. Splitting to the next level: given x with $\text{height}(x) = \alpha$, there are $y, z \in \text{level}_{\alpha+1}(T)$ such that $y \neq z$ and $x <_T y, z$.



3. Extensions at every higher level: $\forall \alpha < \text{height}(T)$, $\forall \beta$ with $\alpha < \beta < \text{height}(T)$, $\forall x \in \text{level}_\alpha(T)$, $\exists y \in \text{level}_\beta(T)$ such that $x <_T y$.
4. Extensionality at limit levels[a]: given a limit ordinal $\alpha < \text{height}(T)$, if $y, z \in \text{level}_\alpha(T)$ and $\{x \mid x <_T y\} = \{x \mid x <_T z\}$, then $y = z$. So the following picture, in which $y \neq z$ sit on a limit level above exactly the same nodes, cannot occur in a normal tree.



[a] so far, in homework, we've only seen trees with at most ω many levels; here, we're not assuming this of every tree

Finally, branches are the paths through a tree, of any length.

Definition 5.2.4 (Branch). A **branch** of a tree $(T, <_T)$ is a subset $b \subset T$ such that $\forall y \in b$, $\forall x <_T y$, $x \in b$ and b is linearly ordered (equivalently, well-ordered) by $<_T$. We say b is an **α-branch** if $\text{type}(b, <_T) = \alpha$.

Two nodes of a tree may or may not lie on a common branch.

Definition 5.2.5 (Comparability and Incomparability). Given a tree T, we say x and y are **comparable** (written $x \parallel y$) iff $x \leq_T y$ or $y \leq_T x$. Else, we say x and y are **incomparable** and write $x \perp y$.

Sets of pairwise incomparable nodes will matter a great deal.

Definition 5.2.6 (Antichain). $A \subseteq T$ is an **antichain** (often abbreviated ac) if $\forall x, y \in A$, $x \neq y$ implies $x \perp y$.

The levels are the basic example.

Example 5.2.7. For every α, $\text{level}_\alpha(T)$ is an antichain.

An antichain may also be as large as possible.

Definition 5.2.8 (Maximal Antichain). An antichain $A \subseteq T$ in a tree T is **maximal** iff

1. A is an antichain
2. There's no antichain B with $B \supsetneq A$

Note: if A is a maximal antichain then for every $x \in T$, there's some $a \in A$ such that $a \parallel x$. Otherwise, $A \cup \{x\} = B$ would be an antichain.

5.3 The Tree Property

We now measure trees against a cardinal: both how tall they are and how wide their levels can be.

Definition 5.3.1 (λ-Tree). Let λ be an infinite cardinal and $(T, <_T)$ be a tree. Then T is a **λ-tree** iff T has height λ and $\forall \alpha < \lambda$, $|\text{level}_\alpha(T)| < \lambda$.

Such a tree is as tall as λ but every level of it is small, and the question is whether it must have a branch running all the way up.

Definition 5.3.2 (Tree Property). λ has the **tree property** (often abbreviated TP) if every normal λ-tree has a λ-branch.

The trees that witness a failure of the tree property have a name.

Definition 5.3.3 (Aronszajn Tree). A λ-**Aronszajn Tree** (often abbreviated λ-A-tree) is a normal λ-tree with no λ-branch.

Thus λ has the TP iff there are no λ-A-trees. At the first two infinite cardinals, the answers go opposite ways.

Theorem 5.3.4 (Kőnig). *ω has the TP.*

One cardinal up, it fails.

Theorem 5.3.5 (Aronszajn). *ω_1 does not have the TP.*

Proofs can be found in Ernest's book [2, Theorems 5.17 and 5.30] and in Kunen [1, Chapter II, Lemma 5.7 and Theorem 5.9].

5.4 The Diamond Principle

We now introduce a principle that guesses every subset of λ correctly on a stationary set of stages. It is what we will use to build a Suslin tree in Section 5.5.

Definition 5.4.1 ($\diamondsuit$). Let λ be a regular, uncountable cardinal. Let $E \subseteq \lambda$. We define $\diamondsuit_\lambda(E)$ to be the statement that there is some sequence $\langle A_\alpha \mid \alpha \in E\rangle$ such that the following both hold:

1. Each A_α is a subset of α
2. For every $X \subseteq \lambda$, the set $S_X = \{\alpha \in E \mid \alpha \cap X = A_\alpha\}$ is a stationary subset of λ.

$\diamondsuit_\lambda(E)$ is shorthand for the statement that there is a $\diamondsuit_\lambda(E)$-sequence, and $\diamondsuit_\lambda$ is shorthand for the statement that there is a $\diamondsuit_\lambda(\lambda)$-sequence.

There are some standard abbreviations.

Notation.

1. $\diamondsuit_\lambda \equiv \diamondsuit_\lambda(\lambda)$
2. $\diamondsuit(E) \equiv \diamondsuit_{\omega_1}(E)$
3. $\diamondsuit \equiv \diamondsuit_{\omega_1}(\omega_1) = \diamondsuit_{\omega_1}$

A $\diamondsuit_\lambda$-sequence guesses every bounded set stationarily often.

Lemma 5.4.2. *Suppose X is a bounded subset of λ (so we have $\beta < \lambda$ such that $X \subseteq \beta$). Assume $\langle A_\alpha \mid \alpha < \lambda \rangle$ is a $\diamondsuit_\lambda$-sequence. Then $\{\alpha < \lambda \mid A_\alpha = X\}$ is stationary in λ.*

This already says something about cardinal arithmetic.

Corollary 5.4.3. $\diamondsuit_\lambda \to 2^{<\lambda} = \lambda$.

In particular, when $\lambda = \omega_1$, we have $2^{<\omega_1} = 2^\omega = \omega_1$, so CH holds. In other words, diamond implies CH.

The sets S_X themselves are nearly disjoint.

Lemma 5.4.4. *Assume $\diamondsuit_\lambda(E)$. Then $\langle S_X \mid X \subseteq \lambda \rangle$ (where S_X is defined from the witness to $\diamondsuit_\lambda(E)$) splits E into 2^λ many almost disjoint pieces. This could mean:*

- *For $X \neq Y$ both subsets of λ, $S_X \cap S_Y$ is bounded in λ*
- *For $X \neq Y$ both subsets of λ, $S_X \cap S_Y$ is non-stationary in λ.*

Proof. For $X \neq Y$, take $\beta = \min(X \triangle Y) < \lambda$. Then, for $\alpha > \beta$, $X \cap \alpha \neq Y \cap \alpha$. So we can't have $X \cap \alpha = A_\alpha$ and $Y \cap \alpha = A_\alpha$. □

Recall the Solovay Splitting Theorem.

Theorem 5.4.5 (Solovay Splitting Theorem). *Given $E \subseteq \lambda$ stationary, with $\lambda = \mathrm{cf}(\lambda) > \omega$, we can partition E into λ many pieces, each of which is stationary.*

5.5 Suslin Trees

We now return to the setting of Suslin's problem (Section 5.1), and to trees of height ω_1.

5.5.1 Characterizing Suslin Trees

sorry

Suslin trees are characterized by their antichains.

Lemma 5.5.1. *$(T, <_T)$ is a Suslin tree iff it's a normal ω_1-tree with no uncountable antichain.*

5.5.2 A Suslin Tree from $\diamondsuit$

We now construct a Suslin tree from a $\diamondsuit$-sequence.

Construction 5.5.2 (Constructing a Suslin Tree from a Diamond). Fix a $\diamondsuit$-sequence $\langle A_\alpha \mid \alpha < \omega_1 \rangle$. By recursion, we'll build trees $(T_\alpha, <_\alpha)$ indexed by ordinals $\alpha < \omega_1$. We'll maintain:

1. T_α is a tree of height α
2. T_α as a set (not as a tree) is a countable ordinal. In fact, $\alpha \leq T_\alpha < \omega_1$
3. For $\alpha < \beta < \omega_1$, T_α is end-extended by T_β (trees) in the "strong sense":
   (a) $<_\alpha \subseteq <_\beta$
   (b) for every $\eta < \alpha$, $\text{level}_\eta(T_\alpha) = \text{level}_\eta(T_\beta)$

The plan is to define $(T, <_T)$ to be the union of the $(T_\alpha, <_\alpha)$s and get that $(T, <_T)$ is a Suslin tree, but other things need to hold for this.

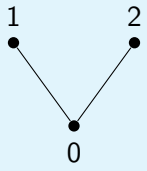
Consider the base case

$$(T_0, <_0) = (0, \emptyset)$$

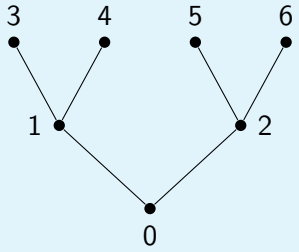
Define $(T_1, <_1) := (\{0\}, \emptyset)$. We want to define T_2 to be the tree with root 0 and children

1 and 2. Or something like that. So define

$$(T_2, <_2) := (\{0, 1, 2\}, \{(0, 1), (0, 2)\})$$

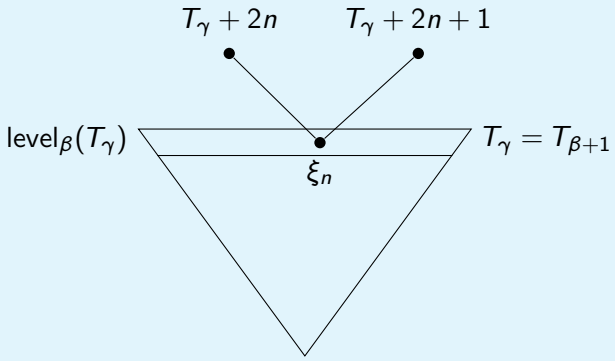


In similar fashion, we define T_3 to be the tree extending T_2 with the children of 1 being 3 and 4 and the children of 2 being 5 and 6.



This is how we define successor trees.

Now consider the successor of a successor level. Say $\delta = \gamma + 1 = (\beta + 1) + 1$. How do we define $(T_\delta, <_\delta)$ based on $(T_\gamma, <_\gamma)$? We know (by the recursion hypothesis) that the universe of T_γ is a countable ordinal. So $\text{level}_\beta(T_\gamma)$ can be listed as ξ_n for $n < \omega$ (for $n < 2^\beta$, if β is finite). Define T_δ by putting $T_\gamma + 2n$ and $T_\gamma + 2n + 1$ above ξ_n.



We see that T_δ (the universe) is $T_\gamma + \omega$ (or $T_\gamma + 2^{\beta+1}$, if β is finite, as with T_2 and T_3).

At every limit ordinal $\beta < \omega_1$, define

$$(T_\beta, <_\beta) := \left(\bigcup_{\alpha<\beta} T_\alpha, \bigcup_{\alpha<\beta} <_\alpha \right)$$

In particular, we now know $(T_\omega, <_\omega)$. It's easy to see that $(T_\omega, <_\omega) \simeq {}^{<\omega}2$.

All that remains is

1. how to define $(T_\delta, <_\delta)$ when $\delta = \gamma + 1$ and γ is a limit ordinal, and
2. why it works.

sorry

Bibliography

These lecture notes are based heavily on the following references:

[1] Kenneth Kunen. *Set Theory: An Introduction to Independence Proofs*. Vol. 102. Studies in Logic and the Foundations of Mathematics. Amsterdam: North-Holland, 1980.

[2] Ernest Schimmerling. *A Course on Set Theory*. Cambridge: Cambridge University Press, 2011. DOI: 10.1017/CBO9780511996351.

Visit https://thefundamentaltheor3m.github.io/SetTheoryNotes/main.pdf for the latest version of these notes. If you have any suggestions or corrections, please feel free to fork and make a pull request to the associated GitHub repository.